

EXPONENTIAL SUMS AND THE RIEMANN ZETA FUNCTION IV

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ABSTRACT

Let $F(x)$ be a real function with sufficiently many derivatives existing and satisfying certain non-vanishing conditions for $1 \leq x \leq 2$. We improve the estimates for the exponential sum

$$S = \sum_m e\left(TF\left(\frac{m}{M}\right)\right),$$

especially when M is close to the square root of T . For the Riemann zeta function,

$$\zeta\left(\frac{1}{2} + iT\right) \ll T^{89/570 + \varepsilon}.$$

We use the 'discrete Hardy–Littlewood method' of Bombieri and Iwaniec. The improvement lies in comparing rational approximations to the derivatives of F at nearby integers m .

1. Introduction

Exponential sums appear when we try to show that two properties are independent on a given sequence of real numbers. We write $[[x]]$ for the nearest integer to the real number x , and $((x)) = x - [[x]]$ with $-\frac{1}{2} < ((x)) \leq \frac{1}{2}$. An important problem of this type is whether $((x_m))$ is distributed uniformly for a given sequence of real numbers $\{x_m\}$. The rounding error in numerical integration can be analysed in terms of a sequence of fractional parts $((x_m))$ as in [7].

This paper continues the work of Bombieri, Iwaniec, Huxley, Watt and Kolesnik [1, 9, 15, 10, 8] on the simple exponential sum

$$S = \sum_{m=M}^{M_2} e\left(TF\left(\frac{m}{M}\right)\right) \quad (1.1)$$

formed with an arbitrary smooth function $F(x)$. Here $e(x)$ denotes $\exp 2\pi ix$, and $M_2 < 2M < T$. There are six steps.

1. Divide the range for m into short intervals, the major and minor arcs. Pick an integer m within each short interval for which the second derivative of $f(x) = TF(x/M)$ is close to a rational number $2a/q$. Approximate $f(m+n)$ by the cubic Taylor polynomial in n .

2. Take a finite Fourier transform modulo q , using Gauss sums.

3. Take a Fourier transform in n . Estimate trivially if q is small (the major arcs).

4. Write the terms of the transformed sum as $e(\mathbf{x} \cdot \mathbf{y})$, where $\mathbf{x} = (h^2, h, h^{3/2}, h^{1/2})$ for some integer h , and $\mathbf{y}$ is constructed from the coefficients of the Taylor polynomial. Use a large sieve inequality to bound the transformed sums in the mean over $\mathbf{y}$.

5. Count the solutions of some inequalities in sums of $\mathbf{x}$ vectors for different integers h .

6. Count the number of 'resonances' between minor arcs, given by the approximate coincidence of y vectors.

There are analogies with the Hardy–Littlewood method treatment of the integral

$$\int_0^1 \left| \sum_{n=1}^N e(f(n + (1 + \alpha)M)) \right|^{10} d\alpha.$$

This paper treats Step 6. The key is the observation by Bombieri and Iwaniec [1] that resonances can be indexed by 'magic matrices' with integer entries and determinant 1, which correspond to coincidences in the first entry y_1 of the y vector, by

$$\begin{pmatrix} a_1 \\ q_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} a \\ q \end{pmatrix}.$$

The third entries, which involve $f^{(3)}(m)$ also, can only coincide for a subinterval of $[M, 2M]$ determined by the magic matrix. In this paper we parametrise the rational numbers a/q in an interval $e/r < a/q < f/s$ by

$$\begin{pmatrix} a \\ q \end{pmatrix} = \begin{pmatrix} f & e \\ s & r \end{pmatrix} \begin{pmatrix} t \\ u \end{pmatrix},$$

where the matrix has integer entries and determinant 1 again. We expand the coefficients of the Taylor series in terms of t and u . In §§ 3 and 4 we show that coincidence is a property of the magic matrix and the minor arc, not depending on the rational value of $f''(x)$ chosen. In § 5 we use this to simplify the coincidence conditions; we use lemmas, proved in § 2, which show that if $u\alpha + t\beta$ is close to an integer for many pairs of integers t, u , then the real numbers α, β are themselves close to integers.

In § 6 we show that if there is a coincidence in all four entries for a long interval of values of x , then the first and third entries coincide to greater accuracy. In §§ 7 and 8 we put all the information together and prove Theorem 1, which gives as a special case

$$S \ll M^{1/2} T^{89/570 + \varepsilon}$$

for any $\varepsilon > 0$ (we explain the order of magnitude symbols below), for M near $T^{1/2}$ under mild conditions on the function $F(x)$. Although we work with all four entries of the minor arc vector y , we do not obtain the full saving possible. We show that among the pairs of minor arcs whose vectors coincide in the first and third entries, at most two thirds of them (on a logarithmic scale) also coincide in the second and fourth entries. Further progress would be possible if one could show that, for most magic matrices, no pair of minor arc vectors coincides in all four entries.

In §§ 9 and 10 we refine the argument under the stronger assumption that $F(x)$ is five times differentiable. The coefficient of n^3 in the Taylor polynomial of Step 1 changes smoothly from one minor arc to the next. This refinement is needed in § 11, where we consider a family of related sums S_i formed with functions $F_i(x) = F(x, y_i)$; the second variable y takes a discrete set of values in $0 \leq y \leq 1$. We require differentiability conditions in y , and extra differentiability in x . Certain combinations of partial derivatives must be bounded away from zero. For instance $F_i(x) = y_i F(x)$ is allowed, but not $F_i(x) = F(x + y_i)$, the 'Weyl shift' of

$F(x)$. Theorem 2 gives a mean fifth power bound for the sums S_i . This yields a small saving on the outer summation in two-variable exponential sums.

As in [10] and [8] we use Theorem 2 and the van der Corput iteration to estimate one-variable sums (1.1) when M is not close to $T^{1/2}$. In § 12 we state the result for M near $T^{1/3}$ in detail as Theorem 3, and summarise results for other ranges as a table of exponents. In Lindelöf's problem on the size of the Riemann zeta function we obtain

$$\zeta(\tfrac{1}{2} + iT) \ll T^{89/570 + \epsilon}.$$

The exponent is 0.1561... The critical range is $M \asymp T^{1/2}$ again, unlike in [8]. The exponent is complicated, and the saving on [8] is small because the argument has several steps in which a number is chosen to give two different terms the same order of magnitude. Each such step complicates the exponents and diminishes the saving.

The bound for the sum (1.1) is expressed in terms of two related parameters N , the length of the short intervals, and R ($\leq N$), the usual size of the denominator q . Treating all arcs as major gives

$$S \ll MT^\epsilon / (NR^2)^{1/6},$$

and in [1] and [9] the bound is

$$S \ll MT^\epsilon / (NR)^{1/4}.$$

In [15] and [10] the bound is

$$S \ll MT^\epsilon / (N^{11}R^9)^{1/40},$$

and in [8],

$$S \ll MT^\epsilon / (N^3R^2)^{1/10}.$$

In this paper

$$S \ll MT^\epsilon / (N^{11}R^4)^{1/30}.$$

Since we subdivide the sum and then take moduli, the limit of the method is

$$S \ll MT^\epsilon / N^{1/2}.$$

We have to take N smaller than in earlier papers, so the improvement is less than one would expect.

This paper has been a long time in gestation. The papers [4] and [5] on integer points close to a curve arose from attempts to count the number of pairs of integers t, u for which the coincidence conditions hold. In this paper we linearise instead. There are points in the argument where the ideas of [5] could still be used, and might improve the result in some cases. The author would like to thank many mathematicians, especially E. Bombieri, C. Hooley, A. Ivic, H. Iwaniec, M. Jutila, and N. Watt for their encouragement. Some of the calculations for § 12 were done on a Spectrum home computer using integer arithmetic within 2^{32} .

NOTATION. Lower case letters apart from x, y, z denote integers. Greek letters denote real numbers. Upper case letters usually denote integers; B_i and C_i are used for constants in inequalities, F, G, H are used as abbreviations for functions $F(x)$, $G(x)$ and $H(t, u)$, O is the order of magnitude sign, S is the exponential

sum of (1.1) and T is the real parameter occurring in it. We also use $e(x)$, $f(x)$, $g(x)$, $h(x)$ for functions of a real variable. In § 2 we use $F(Q)$ to denote the Farey sequence. Section 2 is self-contained and uses its own notation. In other sections we keep the notation of previous papers in this series, and try to avoid using a symbol twice with different meanings. The main construction contains two integer parameters t and u (the denominator and numerator of a rational number), and we write, for example, $b(t, u)$ for the integer b constructed using the values t and u .

We write $e(x)$ for $\exp 2\pi ix$ and use the nearest integer notation

$$x = [[x]] + ((x)),$$

where $[[x]]$ is an integer and $-\frac{1}{2} < ((x)) \leq \frac{1}{2}$. We use $\|x\|$ for $|((x))|$. Note that $((x))$ is the reduction of x modulo 1, and $e(x)$ depends only on $((x))$. We occasionally use $[x]$ for the last integer not exceeding x .

We use the Kloosterman inverse notation: $((\bar{a}/q))$ denotes $((d/q))$ for any integer d with $ad \equiv 1 \pmod{q}$, and the Landau–Vinogradov–Titchmarsh order of magnitude notation: ' $F \ll G$ ' indicates that G is positive and $|F|/G$ is bounded, ' $F \asymp G$ ' means that $F \ll G$ and $G \ll F$, and $O(G)$ stands for some quantity F satisfying $F \ll G$. We use ε for a positive exponent which may be taken arbitrarily close to zero.

2. A linear form often near an integer

Let α and β be real, δ be positive. If α and β are sufficiently close to integers, then

$$\|m\alpha + n\beta\| \leq \delta \tag{2.1}$$

holds for all integers m and n in a box

$$1 \leq m \leq M, \quad 1 \leq n \leq N. \tag{2.2}$$

Conversely, if all integers in the box give solutions of (2.1), then α and β must be close to integers. If we are told only that a large subset of the pairs (m, n) give solutions of (2.1), then other possibilities open. If $\beta = -\alpha$, there are $\min(M, N)$ trivial solutions. If α and β are halves of odd integers, there are at least $\frac{1}{2}(MN - 1)$ solutions.

First we treat the one-dimensional case.

LEMMA 2.1. *Let $n_1, \dots, n_R$ be a subsequence of $1, \dots, N$ with $\|n_r\alpha\| \leq \delta$ for $r = 1, \dots, R$. Then either*

$$R \leq 12\delta N,$$

or, for some $q \leq N/R$,

$$\|q\alpha\| \leq \delta/R,$$

and each integer n_r is a multiple of q .

Proof. By Dirichlet's theorem on Diophantine approximation,

$$|\alpha - a/q| \leq 1/2qN,$$

for some rational number a/q with $q \leq 2N - 1$. Hence

$$\left\| \frac{an_r}{q} \right\| \leq \|n_r \alpha\| + n_r \left| \alpha - \frac{a}{q} \right| \leq \delta + \frac{1}{2q}.$$

Case 1: δq large. If $\delta \geq 1/2q$, then $\|an_r/q\| \leq 2\delta$, and an_r lies in one of e residue classes mod q , where $e = 2[2\delta q] + 1$ and $2\delta q \leq 4\delta q - 1 < e \leq 4\delta q + 1 \leq 6\delta q$. If $R \leq e$, then $R \leq 6\delta q \leq 12\delta N$. If $R \geq e$, then we can write $R = ue + v$, with $u \geq 1$, $0 \leq v \leq e - 1$, so that $N \geq n_r \geq uq$, which gives

$$\frac{R}{N} \leq \frac{ue + v}{uq} \leq \frac{(u+1)e}{uq} \leq \frac{2e}{q} \leq 12\delta,$$

and $R \leq 12\delta N$ again.

Case 2: δq small. If $\delta < 1/2q$, then $\|an_r/q\| < 1/q$, so $q \mid n_r$ and $qR \leq n_r \leq N$. If $\alpha = a/q + \beta$, then $\|n_r \alpha\| = \|n_r \beta\|$, so

$$|qR\beta| \leq \delta, \quad \|q\alpha\| = \|q\beta\| \leq \delta/R.$$

The analogue of Lemma 2.1 for the inequalities (2.1) and (2.2) is the following result. We shall not use this lemma, so the proof is omitted.

LEMMA 2.2. *Suppose that (2.1) and (2.2) have R solutions. Then either*

$$R \leq \max(4M, 4N, 16\delta MN)$$

or there is some integer $s \leq 16MN/9R$ with

$$M \|s\alpha\|, N \|s\beta\| \leq 16\delta MN/9R,$$

and α and β have good rational approximations e/s , f/s , and all solutions of (2.1) and (2.2) have

$$me + nf \equiv 0 \pmod{s}.$$

To remove the denominator s in Lemma 2.2 we suppose that the solutions to (2.1) and (2.2) have a special structure. Let $M = \mu N$ with $\mu \geq 1$, and for some $\lambda > 0$ let S be the set of rational numbers m/n with

$$1 \leq n \leq N, \quad \lambda \leq m/n \leq \mu, \quad (m, n) = 1. \quad (2.3)$$

We suppose that S has R members, and for all members of S , that m and n satisfy (2.1) and (2.2).

LEMMA 2.3. *Suppose that all the R rationals of the set S defined by (2.3) give solutions of (2.1) and (2.2). Then if $N \geq 4$, either*

$$R \leq \max(1932\delta MN, 16N),$$

or α and β are close to integers, with

$$\|\alpha\| \leq 4\delta N/3R, \quad \|\beta\| \leq 161\delta M/R.$$

Proof. For $R \geq 4N$, some denominator n must occur in at least R/N rationals of S . Subtraction in (2.1) gives at least $R/N - 1 \geq 3R/4N$ solutions of $\|m\alpha\| \leq 2\delta$ with $1 \leq m \leq M - 1$. By Lemma 2.1 either $3R/4N \leq 12(2\delta M)$, so $R \leq 32\delta MN$, or $\alpha = a/q + \alpha_1$ with $q \leq M/(3R/4N) = 4MN/3R$, and $|\alpha_1| \leq 4\delta N/3qR$, and for fixed n , all rationals of S have m in a fixed residue class mod q .

To show that $q = 1$ we take n to be a prime p between $\frac{1}{2}N$ and N , so that $(\mu - \lambda)p \geq 3R/4N \geq 12$. The interval $\lambda p \leq m \leq \mu p$ contains at least twelve consecutive integers. Since $p \geq 3$, there must be two consecutive integers m and $m + 1$ which are not multiples of p . Since $m \equiv m + 1 \pmod{q}$, we have $q = 1$. The bound for α becomes $\|\alpha\| \leq 4\delta N/3R$, so that, for any m less than or equal to M ,

$$\|m\alpha\| \leq 4\delta MN/3R.$$

Thus each rational m/n in S has

$$\|n\beta\| \leq \delta + 4\delta MN/3R \leq 7\delta MN/R. \quad (2.4)$$

In order to apply Lemma 2.1 to (2.4) we need to know how many different n occur as denominators. Lemma 9.2 of [6] gives

$$R \geq (\mu - \lambda)[\tfrac{1}{4}N]^2 - N \geq \tfrac{1}{64}(\mu - \lambda)N^2 - \tfrac{1}{16}R,$$

so $\mu - \lambda \leq 68R/N^2$. Each n occurs for at most $(\mu - \lambda)N + 1 \leq 68R/N + 1 \leq 69R/N$ different values of m . Thus at least $\frac{1}{69}N$ values of n occur in (2.4). By Lemma 2.1 either $\frac{1}{69}N \leq 28\delta MN(N/R)$, so $R \leq 1932\delta MN$, or for some $q \leq 69$,

$$\|q\beta\| \leq \frac{697\delta MN}{N} = \frac{161\delta M}{R},$$

and all solutions have n a multiple of q . To see that q must be 1, we observe that the set S must contain a pair of rational numbers m_1/n_1 , m_2/n_2 which are consecutive in some Farey sequence. Then $(n_1, n_2) = 1$, so $q = 1$.

To remove the condition $R \geq 16N$ in Lemma 2.3 we need an explicit lower bound for R in terms of rational approximations to λ and μ . We quote Lemma 9.3 of [6], slightly reworded.

LEMMA 2.4. *Let $F(n)$ denote the Farey sequence of order n . Let e/r and f/s be consecutive fractions in $F(r + s - 1)$, and let J be an interval of length δ lying between e/r and the midpoint $\frac{1}{2}(e/r + f/s)$. For $Q > r$ we have*

$$\sum_{F(16Q) \cap J} 1 \geq \delta Q^2 - 8Q/r.$$

LEMMA 2.5. *Suppose that all the R rationals of the set S defined by (2.3) give solutions of (2.1) and (2.2), and for some $B \geq 1$,*

$$R \geq \max((\mu - \lambda)N^2/B, 2). \quad (2.5)$$

If

$$M\delta \leq 1/96B, \quad (2.6)$$

then α and β are both integers, with

$$\|\alpha\| \leq 8N\delta/R, \quad \|\beta\| \leq 8M\delta/R.$$

Proof. We number the members of S as m_r/n_r in order of n_r increasing. The determinants

$$D_r = \begin{vmatrix} m_R & m_r \\ n_R & n_r \end{vmatrix}$$

are all different, since $D_r = D_s$ implies that

$$m_R(n_r - n_s) = n_R(m_r - m_s),$$

and $(m_R, n_R) = 1$ with $|n_r - n_s| \leq n_R - 1$, so that n_r must equal n_s , and then m_r equals m_s . There are $R - 1$ non-zero determinants, of which at least $\frac{1}{2}(R - 1)$ have the same sign. We have

$$|D_r| \leq (\mu - \lambda)n_r n_R \leq (\mu - \lambda)N^2 \leq BR,$$

and

$$\begin{aligned} \|D_r \alpha\| &= \|n_r(m_R \alpha + n_R \beta) - n_R(m_r \alpha + n_r \beta)\| \\ &\leq (n_r + n_R)\delta \leq 2N\delta. \end{aligned}$$

Lemma 2.1 now gives two possibilities. Either

$$\frac{1}{2}(R - 1) \leq 12BR \cdot 2N\delta, \quad (2.7)$$

which gives $R \leq 96BRN\delta$, or for some q ,

$$\|q\alpha\| \leq \frac{2N\delta}{\frac{1}{2}(R - 1)} \leq \frac{8N\delta}{R},$$

and q is a factor of each determinant D_r . Some fraction m_r/n_r must be adjacent to m_R/n_R in the Farey sequence $F(n_R)$, so that $D_r = \pm 1$, and q must be 1.

Similarly we find that $\|D_r \beta\| \leq 2M\delta$, and Lemma 2.1 gives either

$$\frac{1}{2}(R - 1) \leq 12BRM\delta, \quad (2.8)$$

or for some q (which again must be unity as above)

$$\|q\beta\| \leq 8M\delta/R.$$

The condition (2.6) contradicts (2.7) and (2.8).

Finally we combine Lemmas 2.3 and 2.5. The constants in Lemma 2.6 can scarcely be best possible, but that is not important right now.

LEMMA 2.6. *Suppose that the set S defined by (2.3) has R elements, and for some $B \geq 1$,*

$$R \geq \max((\mu - \lambda)N^2/B, 2),$$

and suppose that all the elements of S give solutions of (2.1) and (2.2). Then either

$$R \leq 1536B\delta MN, \quad (2.9)$$

or α and β are close to integers, with

$$\|\alpha\| \leq 8\delta N/R, \quad \|\beta\| \leq 161\delta M/R.$$

Proof. If $\delta M \leq 1/96B$, then Lemma 2.5 applies. If δM is larger than this bound, but (2.9) is false, then $R > 16N$, and Lemma 2.3 applies.

3. Parametrising rational points

The exponential sum of length M is divided into short intervals (the major and minor arcs) of length about N . On each short interval we pick a rational number

a/q which is a value of $\frac{1}{2}f''(x)$, then an integer m with

$$\frac{1}{2}f''(m) = a/q + \lambda, \quad \lambda \ll \max |f^{(3)}(x)| \asymp 1/NR^2;$$

this inequality introduces the parameter R which is also important in the construction. For this m we put

$$qf'(m) = c + \delta, \quad -\frac{1}{2} < \delta \leq \frac{1}{2}, \quad \mu = \frac{1}{6}f^{(3)}(m), \quad \nu = \frac{1}{24}f^{(4)}(m).$$

The short exponential sums on the minor arcs are transformed by Poisson summation. Different minor arcs may give rise to sums which are very similar, differing only in their endpoints or in a possible phase factor. We call this 'coincidence' or 'resonance' between minor arcs. There are different cases according to whether q is odd or even, but multiplying (5.4) and (5.5) in [9] by 4 and by 2 respectively gives the necessary conditions for a coincidence with q in the range $Q \leq q < 2Q$ in the common form

$$\left\| \frac{\bar{a}}{q} - \frac{\bar{a}_1}{q_1} \right\| \leq \Delta_1 \asymp \frac{R^4}{N^2 Q^2}, \quad (3.1)$$

$$\left\| \frac{\bar{a}c}{q} - \frac{\bar{a}_1 c_1}{q_1} \right\| \leq \Delta_3 \asymp \frac{R^2}{NQ}, \quad (3.2)$$

$$\left| \frac{\mu_1 q_1^3}{\mu q^3} - 1 \right| \leq \Delta_2 \asymp \frac{R^2}{N^2}, \quad (3.3)$$

$$|\delta - \delta_1| \leq \Delta_4 \asymp \frac{Q}{N}; \quad (3.4)$$

the variables with suffix 1 refer to the other minor arc. The construction ensures that $R \ll Q \ll N$, so that $\Delta_1 \ll \Delta_2 \ll \Delta_3 \ll \Delta_4$. We shall refer to (3.1) to (3.4) as the First, Second, Third and Fourth Conditions.

Bombieri and Iwaniec [1] observed that the First Condition (3.1) is equivalent to

$$\begin{pmatrix} a_1 \\ q_1 \end{pmatrix} = M \begin{pmatrix} a \\ q \end{pmatrix}, \quad M = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where M is an integer matrix of determinant 1, and that if q has size about Q , then

$$C \ll \Delta_1 Q^2 \ll R^4/N^2, \quad B \ll \frac{a^2}{q^2} \Delta_1 Q^2 + \frac{|a|}{q}, \quad A, D \ll \frac{|a|}{q} C + 1.$$

The set of possible magic matrices is thus independent of the order of magnitude Q .

The identity matrix corresponds to trivial coincidences $a_1/q_1 = a/q$. For other matrices the Third Condition (3.3) may restrict a/q to a short interval; this fact was the main step in the proof of Lemma 2.4 in [9]. We consider intervals with endpoints $e/r, f/s$, consecutive fractions in some Farey sequence, with

$$fr - es = 1. \quad (3.5)$$

The rationals a/q in the interval can be written as

$$a = eu + ft, \quad q = ru + st$$

with $(t, u) = 1$. Writing

$$M \begin{pmatrix} f & e \\ s & r \end{pmatrix} = \begin{pmatrix} f_1 & e_1 \\ s_1 & r_1 \end{pmatrix},$$

we have

$$\begin{pmatrix} a_1 \\ q_1 \end{pmatrix} = M \begin{pmatrix} eu + ft \\ ru + st \end{pmatrix} = \begin{pmatrix} e_1u + f_1t \\ r_1u + s_1t \end{pmatrix}.$$

We suppose that the Coincidence Conditions (3.2), (3.3) and (3.4) hold at a/q , but not necessarily at e/r or at f/s . Since $(t, u) = 1$, there are integers $\bar{t}, \bar{u}$ with

$$t\bar{t} + u\bar{u} = 1. \quad (3.6)$$

We note the identities

$$(eu + ft)(r\bar{t} - s\bar{u}) - (e\bar{t} - f\bar{u})(ru + st) = 1 \quad (3.7)$$

and

$$\frac{r\bar{t} - s\bar{u}}{ru + st} + \frac{\bar{u}}{t} = \frac{r}{t(ru + st)}. \quad (3.8)$$

We think of a/q and the coefficients of the Taylor polynomial as functions of t and u , so we write $c(t, u)$, $\delta(t, u)$, $\lambda(t, u)$, $\mu(t, u)$, $\nu(t, u)$ and $m(t, u)$ for the quantities corresponding to a/q , with suffix 1 for those constructed from a_1/q_1 . We use the letters $c, m, \delta, \lambda, \mu, \nu$ (and these only) without argument for the values at $t=0, u=1$, which occur frequently in the formulae that follow. We introduce a new variable by

$$n = m(t, u) - m, \quad n_1 = m_1(t, u) - m_1.$$

How n and n_1 depend on t and u will be elucidated in (3.11) below. We take Taylor series about m and m_1 with n and n_1 as the variables of expansion. These expansions cover U consecutive minor arcs, with

$$U \leq R^2/rs, \quad n \ll NU. \quad (3.9)$$

The Taylor series given are about the left-hand endpoint; similar expansions are possible about the right-hand endpoint, with sign changes which are left to the curious reader.

At this stage we make the order of magnitude assumptions

$$f^{(3)}(x) \asymp 1/NR^2, \quad f^{(4)}(x) \ll |f^{(3)}(x)|/M \ll 1/MNR^2,$$

uniformly on the interval $(1 - \delta)M \leq x \leq (2 + \delta)M$ for some $\delta > 0$. These assumptions guide us in which terms to keep, and which terms to treat as perturbations. The mean value theorem in the form

$$f''(m + n) = f''(m) + nf^{(3)}(m) + O(n^2 \max |f^{(4)}(x)|)$$

gives

$$\frac{2(eu + ft)}{(ru + st)} + 2\lambda(t, u) = \frac{2e}{r} + 2\lambda + 6n\mu + O\left(\frac{n^2}{MNR^2}\right),$$

so that by (3.5),

$$3n\mu\left(1 + O\left(\frac{n}{M}\right)\right) = \frac{t}{r(ru + st)} + \lambda(t, u) - \lambda. \quad (3.10)$$

Approximating further, we have

$$n = \frac{t}{3\mu r(ru + st)} + O\left(1 + \frac{n^2}{M}\right) = G\left(\frac{u}{t}\right) + O\left(1 + \frac{n^2}{M}\right), \quad (3.11)$$

where $G(x)$ is the continuous function

$$G(x) = 1/3\mu r(rx + s). \quad (3.12)$$

We make constant use of (3.11) to pass from the discrete to the continuous.

The mean value theorem for the first derivative gives

$$f'(m + n) = f'(m) + nf''(m) + \frac{1}{2}n^2f^{(3)}(m) + O(n^3 \max |f^{(4)}(x)|),$$

so that

$$\begin{aligned} \frac{c(t, u) + \delta(t, u)}{ru + st} &= \frac{c + \delta}{r} + \frac{2en}{r} + 2n\lambda + 3n^2\mu + O\left(\frac{n^3}{MNR^2}\right) \\ &= \frac{c + \delta}{r} + \frac{2en}{r} + n\lambda(t, u) + n\lambda + \frac{nt}{r(ru + st)} + O\left(\frac{n^3}{MNR^2}\right) \\ &= \frac{c + \delta}{r} + \frac{2en}{r} + \frac{nt}{r(ru + st)} + \frac{\gamma}{ru + st}, \end{aligned} \quad (3.13)$$

say, where we have used (3.10) to eliminate the non-arithmetic term in μ . The non-arithmetic residual term γ in (3.13) depends on t and u . We analyse γ later. In order of magnitude we have

$$\begin{aligned} \frac{\gamma}{ru + st} &\ll \frac{n}{NR^2} + \frac{n^3}{MNR^2} \\ &\ll \left(1 + \frac{n^2}{M}\right) \left(\frac{t}{r(ru + st)} + \frac{1}{NR^2} \left(1 + \frac{n^2}{M}\right)\right) \end{aligned} \quad (3.14)$$

by (3.11). We shall see that γ is negligible when the right-hand side of (3.14) is $\ll 1/N$.

We have

$$\begin{aligned} c(t, u) + \delta(t, u) &= \frac{(ru + st)(c + 2en)}{r} + \frac{\delta(ru + st)}{r} + \frac{nt}{r} + \gamma \\ &= (c + 2en)u + \frac{cst}{r} + \frac{2nt}{r}(fr - 1) + \frac{\delta(ru + st)}{r} + \frac{nt}{r} + \gamma \\ &= cu + 2n(eu + ft) + H, \end{aligned} \quad (3.15)$$

where

$$H = H(t, u) = \frac{(cs - n)t}{r} + \frac{\delta(ru + st)}{r} + \gamma. \quad (3.16)$$

Recalling the notation $[[x]]$, $((x))$, we put

$$\begin{aligned} h &= h(t, u) = [[H(t, u)]], \\ \eta &= \eta(t, u) = ((H(t, u))). \end{aligned}$$

Then

$$\begin{aligned} c(t, u) &= cu + 2n(eu + ft) + h, \\ \delta(t, u) &= \eta. \end{aligned}$$

By (3.7) we have

$$\begin{aligned} (r\bar{t} - s\bar{u})c(t, u) &= cu(r\bar{t} - s\bar{u}) + 2n(eu + ft)(r\bar{t} - s\bar{u}) + (r\bar{t} - s\bar{u})h \\ &= c(ru\bar{t} + st\bar{t} - s) + 2n(ru + st)(e\bar{t} - f\bar{u}) + 2n + (r\bar{t} - s\bar{u})h. \end{aligned}$$

We are now ready for the subtle calculation at the heart of this paper: by (3.8),

$$\begin{aligned} \left(\left(\frac{r\bar{t} - s\bar{u}}{ru + st} c(t, u) \right) \right) &= \left(\left(\frac{2n - cs}{ru + st} + \frac{r\bar{t} - s\bar{u}}{ru + st} h(t, u) \right) \right) \\ &= \left(\left(\frac{2n - cs}{ru + st} + \left(\frac{r}{t(ru + st)} - \frac{\bar{u}}{t} \right) h \right) \right) \\ &= \left(\left(\frac{2n - cs}{ru + st} + \frac{r(H - \eta)}{t(ru + st)} - \frac{\bar{u}h}{t} \right) \right) \\ &= \left(\left(\frac{2n - cs}{ru + st} + \frac{cs - n}{ru + st} + \frac{\delta}{t} + \frac{r(\gamma - \eta)}{t(ru + st)} - \frac{\bar{u}h}{t} \right) \right) \\ &= \left(\left(\frac{n}{ru + st} + \frac{\delta}{t} - \frac{\bar{u}h}{t} + \frac{r(\gamma - \eta)}{t(ru + st)} \right) \right). \end{aligned}$$

We subtract two expressions of this type to get the Second Condition (3.2). The Fourth Condition (3.4) in this notation is

$$|\eta - \eta_1| \ll Q/N,$$

so we can suppose that $|\eta - \eta_1|$ is less than $\frac{1}{4}$, and that H and H_1 are approximately congruent modulo 1. Thus

$$[[H_1 - H]] = [[H_1]] - [[H]] = h_1 - h.$$

When (3.4) holds, the left-hand side of (3.2) is

$$\begin{aligned} &\left\| \frac{n}{ru + st} - \frac{n_1}{(r_1u + s_1t)} + \frac{\delta - \delta_1}{t} - \frac{\bar{u}(h - h_1)}{t} \right. \\ &\quad \left. + \frac{\eta_1 - \gamma_1}{t} \left(\frac{r_1}{r_1u + s_1t} - \frac{r}{ru + st} \right) + \frac{r}{t(ru + st)} (\eta_1 - \gamma_1 - \eta + \gamma) \right\|. \quad (3.17) \end{aligned}$$

4. Coincidence over a short interval

In this section we show that coincidence is a property of the minor arc, not of the particular rational point a/q chosen, in the sense that if the Coincidence Conditions hold at e/r , then they hold at $(eu + ft)/(ru + st)$ whenever $n(t, u) \ll$

N , but possibly weakened by a constant factor. We assume that

$$\frac{s_1}{r_1} - \frac{s}{r} = \frac{C}{rr_1} \ll \frac{1}{r^2} \frac{R^4}{N^2} \ll 1, \quad (4.1)$$

$$\left\| \frac{cs}{r} - \frac{c_1 s_1}{r_1} \right\| \ll \frac{R^2}{rN} \ll 1, \quad (4.2)$$

$$\frac{\mu_1 r_1^3}{\mu r^3} - 1 \ll \frac{R^2}{N^2}, \quad (4.3)$$

$$\delta - \delta_1 \ll \frac{r}{N}. \quad (4.4)$$

We note that by (3.11), $tR^2/rQ \approx n/N \ll 1$. We examine the Fourth Condition. We have

$$\begin{aligned} ((\eta - \eta_1)) = & \left(\left(t \left(\frac{cs}{r} - \frac{c_1 s_1}{r_1} \right) - t \left(\frac{n}{r} - \frac{n_1}{r_1} \right) + \frac{ru + st}{r} (\delta - \delta_1) \right. \right. \\ & \left. \left. + \delta_1 \left(\frac{ru + st}{r} - \frac{r_1 u + s_1 t}{r_1} \right) + \gamma - \gamma_1 \right) \right). \end{aligned} \quad (4.5)$$

By (4.2),

$$\left\| t \left(\frac{cs}{r} - \frac{c_1 s_1}{r_1} \right) \right\| \leq t \left\| \frac{cs}{r} - \frac{c_1 s_1}{r_1} \right\| \ll \frac{tR^2}{rN} \ll \frac{tR^2}{rQ} \Delta_4.$$

Next we note that δ_1 is bounded, and by (4.1),

$$\frac{ru + st}{r} - \frac{r_1 u + s_1 t}{r_1} \ll \frac{t}{r^2} \frac{R^4}{N^2} \ll \frac{tR^2}{rQ} \Delta_4. \quad (4.6)$$

By (3.14),

$$\gamma - \gamma_1 \ll \frac{nQ}{NR^2} \left(1 + \frac{n^2}{M} \right) \ll \frac{tR^2}{rQ} \Delta_4 \frac{N}{R^2} \left(1 + \frac{N^2}{M} \right) \ll \frac{tR^2}{rQ} \Delta_4, \quad (4.7)$$

provided that

$$N^3 \ll MR^2, \quad N \ll MT^{-1/4}, \quad (4.8)$$

which will be an assumption of the method.

To discuss the second term on the right of (4.5) we put

$$g(x) = \frac{1}{3\mu r^2(rx + s)} - \frac{1}{3\mu_1 r_1^2(r_1 x + s_1)}, \quad (4.9)$$

so that

$$\begin{aligned} g^{(k)}(x) &= \frac{(-1)^k r^{k-2} k!}{3\mu (rx + s)^{k+1}} - \frac{(-1)^k r_1^{k-2} k!}{3\mu_1 (r_1 x + s_1)^{k+1}} \\ &= \frac{(-1)^k k!}{3\mu r^3} \frac{r_1^{k+1}}{(r_1 x + s_1)^{k+1}} \left(\left(\frac{s_1/r_1 + x}{s/r + x} \right)^{k+1} - \frac{\mu r^3}{\mu_1 r_1^3} \right). \end{aligned} \quad (4.10)$$

We obtain the bound for $g^{(k)}(x)$ under more general conditions than we need in this section, since we shall use it later. Suppose also that the Third Condition (3.3) holds at a/q , so that

$$\frac{\mu_1(t, u)(r_1 u + s_1 t)^3}{\mu(t, u)(ru + st)^3} - 1 \ll \frac{R^2}{N^2}.$$

By the bound for $f^{(4)}(x)$, $\mu(t, u)/\mu = 1 + O(n/M)$, and similarly for $\mu_1(t, u)$, so that

$$\frac{r_1 u + s_1 t}{ru + st} = \frac{r_1}{r} \left(1 + O\left(\frac{R^2}{N^2} + \frac{n}{M}\right) \right), \quad (4.11)$$

and for $x = u/t$,

$$g^{(k)}\left(\frac{u}{t}\right) \ll \frac{NR^2}{r^3} \left(\frac{rt}{Q}\right)^{k+1} \left(\frac{R^2}{N^2} + \frac{n}{M}\right). \quad (4.12)$$

For $n \ll N$, (4.8) gives

$$tg\left(\frac{u}{t}\right) \ll \frac{R^4}{r^3 N} \frac{rt^2}{Q} \ll \left(\frac{tR^2}{rQ}\right)^2 \Delta_4.$$

By (3.11),

$$t\left(\frac{n}{r} - \frac{n_1}{r_1}\right) = tg\left(\frac{u}{t}\right) + O\left(\left(1 + \frac{n^2}{M}\right) \frac{t}{r}\right) \ll \frac{tR^2}{rQ} \Delta_4.$$

Thus (4.5) gives

$$((\eta - \eta_1)) = \left(\left(\frac{ru + st}{r} (\delta - \delta_1) \right) \right) + O\left(\frac{tR^2}{rQ} \Delta_4\right).$$

We would like to deduce that $\eta - \eta_1 \ll \Delta_4$, so that the Fourth Condition (3.4) holds at $a/q = (eu + ft)/(ru + st)$. However, it is possible that $\eta = \delta(t, u)$ is close to $\frac{1}{2}$, and $\eta_1 = \delta_1(t, u)$ is close to $-\frac{1}{2}$. To get round this, we modify the construction so that whenever $\delta(t, u) = \pm\frac{1}{2} + O(\Delta_4)$, then we consider two values of $c(t, u)$ differing by unity, with the corresponding values of $\delta(t, u)$ close to $\frac{1}{2}$ and to $-\frac{1}{2}$. For two of the four choices of $c(t, u)$ and $c_1(t, u)$, there is a coincidence in the Fourth Condition at $a/q = (eu + ft)/(ru + st)$. In effect we are counting these minor arcs twice, increasing the final estimate for the exponential sum.

For some integer b ,

$$cs/r - c_1s_1/r_1 = b + O(R^2/rN).$$

Thus

$$H - H_1 = bt + \frac{ru + st}{t} (\delta - \delta_1) + O\left(\frac{tR^2}{rQ} \Delta_4\right) = bt + O(\Delta_4),$$

and $h - h_1 = bt$, with $\bar{u}(h - h_1)/t = \bar{u}b$, an integer. Also

$$\eta - \eta_1 = \frac{ru + st}{r}(\delta - \delta_1) + O\left(\frac{tR^2}{rN}\right).$$

The left-hand side of (3.17) is thus

$$\left\| \frac{n}{ru + st} - \frac{n_1}{r_1u + s_1t} + (\eta_1 - \gamma_1) \left(\frac{r_1}{t(r_1u + s_1t)} - \frac{r}{t(ru + st)} \right) + \frac{r(\gamma - \gamma_1)}{t(ru + st)} \right\| + O\left(\frac{r}{tQ} \frac{tR^2}{rN}\right), \quad (4.13)$$

the error term being $O(\Delta_3)$.

By (4.7),

$$\frac{r(\gamma - \gamma_1)}{t(ru + st)} \ll \frac{r}{tQ} \frac{tR^2}{rN} \ll \Delta_3.$$

Also $\eta_1 - \gamma_1$ is bounded, and as in (4.6),

$$(\eta_1 - \gamma_1) \left(\frac{r_1}{t(r_1u + s_1t)} - \frac{r}{t(ru + st)} \right) \ll \frac{C}{Q^2} \ll \Delta_1 \ll \Delta_3.$$

Finally by (3.11), (4.11), and (4.8),

$$\begin{aligned} \frac{n}{ru + st} - \frac{n_1}{r_1u + s_1t} &= -\frac{1}{t} g'\left(\frac{u}{t}\right) + O\left(\frac{1}{Q} \left(1 + \frac{n^2}{M}\right)\right) \\ &\ll \frac{R^4}{NQ^2} \frac{t}{r} + \frac{R^2}{NQ} \ll \Delta_3. \end{aligned}$$

Thus the Second Condition (3.2) continues to hold at a/q , possibly weakened by a constant factor. We know from the analysis of [9] that, apart from a constant factor, the Third Condition (3.3) holds for each a/q representing a particular minor arc if it holds for any one of them. Apart from constant factors, the Coincidence Conditions (3.1) to (3.4) hold throughout the minor arc corresponding to e/r .

5. Linearising the fourth condition

After § 4 we can suppose that the Coincidence Conditions (3.1) to (3.4) hold for all a/q which are values of $\frac{1}{2}f''(x)$ on a particular minor arc, but with the inequalities weakened from $\leq \Delta_i$ to $\ll \Delta_i$. Our object is to remove the term in $\bar{u}$ from (3.17) and to replace the absolute value modulo 1 by an ordinary absolute value. We make a delicate approximation in the Fourth Condition (3.4) and use the results of § 2.

We use the notation of § 3, so that

$$Q \leq q = ru + st < 2Q.$$

The rational values of $\frac{1}{2}f''(x)$,

$$\frac{a}{q} = \frac{eu + ft}{ru + st} = \frac{e}{r} + \frac{1}{r(ru/t + s)},$$

lie in an interval of length $\asymp 1/R^2$, which corresponds to an interval in $x = u/t$ of length $\asymp (r^2x^2 + s^2)/R^2$ for $t \gg rQ/R^2$, $n \gg N$. We take $n \ll NU$, where $U \leq R^2/rs$. By (3.11),

$$t \asymp nrQ/NR^2 \ll rQU/R^2.$$

As in (4.5), the Fourth Condition (3.4) is

$$\left\| t \left(\frac{cS}{r} - \frac{c_1s_1}{r_1} \right) - t \left(\frac{n}{r} - \frac{n_1}{r_1} \right) + \frac{ru + st}{r} \delta - \frac{r_1u + s_1t}{r_1} \delta_1 + \gamma - \gamma_1 \right\| \ll \Delta_4. \quad (5.1)$$

Three of the five terms on the left of (5.1) are linear in t and u . We combine the other two terms. By (3.13) we have

$$\begin{aligned} \frac{\gamma(t, u)}{ru + st} &= \frac{c(t, u) + \delta(t, u)}{ru + st} - \frac{c + \delta}{r} - \frac{2en}{r} - \frac{nt}{r(ru + st)} \\ &= 2n\lambda + 3n^2\mu - \frac{nt}{r(ru + st)} + O\left(\frac{n^3}{MNR^2}\right) \\ &= n(2\lambda + 3\mu(n - G)) + O(n^3/MNR^2), \end{aligned} \quad (5.2)$$

where we have written G for $G(u/t)$. The error term in (5.2) is $O(1/N)$ for

$$n \ll (MR^2)^{1/3}. \quad (5.3)$$

We use (3.11) repeatedly to get

$$\frac{\gamma(t, u)}{ru + st} = (2\lambda + 3\mu(n - G))G + O\left(\frac{1}{N}\right).$$

Thus

$$\begin{aligned} \gamma(t, u) &= \frac{t}{3\mu r} (2\lambda + 3\mu(n - G)) + O\left(\frac{Q}{N}\right) \\ &= \frac{t}{r} \left(n - G + \frac{2\lambda}{3\mu} \right) + O(\Delta_4). \end{aligned}$$

When we substitute into (5.1), then the only non-linear term is

$$-\frac{t}{r} G\left(\frac{u}{t}\right) + \frac{t}{r_1} G_1\left(\frac{u}{t}\right) = -tg\left(\frac{u}{t}\right).$$

We write (5.1) as

$$\|\alpha u + \beta t - tg(u/t)\| \ll \Delta_4 \quad (5.4)$$

with

$$\alpha = \delta - \delta_1, \quad (5.5)$$

$$\beta = \frac{(c + \delta)s}{r} - \frac{(c_1 + \delta_1)s_1}{r_1} + \frac{2\lambda}{3\mu r} - \frac{2\lambda_1}{3\mu_1 r_1}. \quad (5.6)$$

Let x_0 be a value of u/t on the minor arc. We take a Taylor approximation to $g(x_0 + y)$ with $y \ll (r^2x^2 + s^2)/R^2$. In the remainder term we have, for some θ

in $0 < \theta < 1$,

$$\begin{aligned} ty^2 g''(x_0 + \theta y) &< \frac{t(r^2 x^2 + s^2)^2}{R^4} \frac{R^4}{r^3 N} \left(\frac{rt}{Q}\right)^3 \\ &< t^4 \frac{Q^4}{t^4} \frac{1}{NQ^3} < \Delta_4 \end{aligned}$$

when (5.3) holds. For values of u/t corresponding to the minor arc under consideration we have a linear approximation of the form

$$H - H_1 = \alpha' u + \beta' t + O(\Delta_4),$$

with

$$\alpha' = \alpha - g'(x_0), \quad (5.7)$$

and

$$\beta' = \beta - g(x_0) + x_0 g'(x_0). \quad (5.8)$$

Then $\|\alpha' u + \beta' t\| \leq \delta' < \Delta_4$.

Let a_0/q_0 be the rational number of smallest denominator which is a value of $\frac{1}{2}f''(x)$ on the minor arc. By the analysis of § 9 of [6], for

$$Q \geq B_1 \max(q_0, R^2/q_0) \quad (5.9)$$

there are R' rational values a/q of $\frac{1}{2}f''(x)$ with $q < 2Q$ on the minor arc, and $R' \geq Q^2/B_2 R^2$. Here B_1 and B_2 are certain constants. The corresponding integers t and u lie in ranges

$$t \leq N' \asymp rQG(u/t)/NR^2, \quad u \leq M' \asymp Q/r, \quad M' = \mu' N',$$

and u/t lies in an interval of length

$$\mu' - \lambda' \asymp \frac{r^2 x^2 + s^2}{R^2} \asymp \frac{Q^2}{t^2 R^2} \asymp \frac{R'^2}{N'^2}.$$

We have set up the notation and conditions of Lemma 2.6 with parameters M' , N' , R' and δ' with some bounded B' . The dashes distinguish the parameters of Lemma 2.6 from the usual meanings of the symbols M , N , etc. We suppose that $M' \geq N'$; there is a similar bound if $N' \geq M'$. Either $R' \leq 1536B'\delta'M'N'$, which is impossible if

$$G(u/t) < N^2/B_3 Q \quad (5.10)$$

for some $B_3 \geq 1$, or α' and β' are close to integers, with

$$\|\alpha'\| \leq 8\delta'N'/R' < rG/N^2, \quad (5.11)$$

and

$$\|\beta'\| \leq 161\delta'M'/R' < R^2/rN. \quad (5.12)$$

Let $a = [[\alpha']]$, $b = [[\beta']]$. When (5.11) and (5.12) hold, then

$$|\alpha' u + \beta' t - au - bt| \leq \frac{169\delta'M'N'}{R'} < \frac{169}{1536B'} < \frac{1}{2}.$$

Thus we have

$$h - h_1 = [[H - H_1]] = au + bt.$$

We can divide by t in (5.4) to get

$$\left\| (\alpha - a) \frac{u}{t} + \beta - g\left(\frac{u}{t}\right) \right\| \ll \frac{\Delta_4}{t} \ll \frac{R^2}{rG}. \quad (5.13)$$

In (3.17) we have

$$\frac{\bar{u}}{t}(h - h_1) = \frac{a(1 - t\bar{i})}{t} + b\bar{u} = \frac{a}{t} - a\bar{i} + b\bar{u}. \quad (5.14)$$

By (4.8), (4.11) gives

$$\begin{aligned} \frac{\eta_1}{t} \left(\frac{r_1}{r_1 u + s_1 t} - \frac{r}{ru + st} \right) &\ll \frac{1}{t} \left| \frac{r_1}{r_1 u + s_1 t} - \frac{r}{ru + st} \right| \\ &\ll \frac{r}{Q} \left(\frac{R^2}{N^2} + \frac{n}{tM} \right) \ll \left(\frac{r}{N} + \frac{N^2}{Mq} \right) \Delta_3 \ll \Delta_3. \end{aligned}$$

When the Fourth Condition (3.4) holds, then

$$\frac{r(\eta_1 - \eta)}{t(ru + st)} \ll \frac{r}{tQ} \frac{Q}{N} \ll \frac{R^2}{NQ} \asymp \Delta_3.$$

For

$$n^2 \ll MR^2/N \quad (5.15)$$

(another consequence of (5.3)), we have

$$\frac{\gamma r}{t(ru + st)} \ll \frac{r}{Q} \left(1 + \frac{n^2}{M} \right) \left(\frac{1}{r} + \frac{ru + st}{NR^2} \left(1 + \frac{n^2}{M} \right) \right) \ll \Delta_3.$$

The Second Condition in the form (3.17) has now been simplified to

$$\left\| \frac{n}{ru + st} - \frac{n_1}{r_1 u + s_1 t} + \frac{\delta - \delta_1 - a}{t} \right\| \ll \Delta_3. \quad (5.16)$$

We note that by (3.11), (4.10) and (4.12) we have

$$\begin{aligned} \left| \frac{n}{ru + st} - \frac{n_1}{r_1 u + s_1 t} \right| &= \left| -\frac{1}{t} g'\left(\frac{u}{t}\right) + O\left(\frac{1}{Q} \left(1 + \frac{n^2}{M}\right)\right) \right| \\ &\leq B_4 \frac{G}{N} \frac{R^2}{NQ} \leq \frac{B_4}{B_3} \frac{R^2}{Q^2} \leq \frac{B_4}{B_1^2 B_3} < \frac{1}{8} \end{aligned}$$

for some absolute constant B_4 , if we have chosen B_1 sufficiently large. Since $a = [[\alpha - g'(x_0)]]$, by (4.12) we have

$$a \ll 1 + \frac{NR^2}{r^3} \left(\frac{rt}{Q} \right)^2 \frac{R^2}{N^2} \ll 1 + \frac{n^2 r}{N^3}. \quad (5.17)$$

Then

$$\left| \frac{\delta - \delta_1 - a}{t} \right| \ll \frac{1}{t} + \frac{|n| R^2}{N^2 Q},$$

and $|(\delta - \delta_1 - a)/t| < \frac{1}{8}$ if B_1 and B_2 are sufficiently large, and (5.10) holds. Thus the expression on the left of (5.16) is the absolute value modulo 1 of a number

that is numerically less than $\frac{1}{2}$, which is the ordinary absolute value. We deduce that

$$\left| \frac{\delta - \delta_1 - a}{t} - \frac{1}{t} g' \left(\frac{u}{t} \right) \right| \ll \Delta_3,$$

and multiplying by t gives

$$\left| \alpha - a - g' \left(\frac{u}{t} \right) \right| \ll \frac{tR^2}{NQ} \ll \frac{Gr}{N^2}. \quad (5.18)$$

6. Coincidence over a long interval

In § 3 we set up the parametrisation of the rational values of $\frac{1}{2}f''(x)$ in general. So far we have considered only one minor arc at a time. We know from § 5 that if the Coincidence Conditions (3.1) to (3.4) hold on a particular minor arc, and if (5.9) and (5.10) hold, then the Fourth and Second Conditions (3.4) and (3.2) simplify to (5.13) and (5.18), which we write as

$$|(\alpha - a)x + \beta + b - g(x)| \leq B_5 R^2 / rG(x), \quad (6.1)$$

$$|\alpha - a - g'(x)| \leq B_6 rG(x) / N^2, \quad (6.2)$$

where $x = u/t$, B_5 and B_6 are constants. We consider the range of x for which the bounds on the right of (6.1) and (6.2) are less than $\frac{1}{4}$.

First we consider the possible integer values of a and b . We define a new function $k(y)$ implicitly by

$$g'(k(y)) = \alpha - y,$$

and we put

$$z = K(y) = (y - \alpha)k(y) + g(k(y)) - \beta.$$

Then $k(y)$ and $K(y)$ are algebraic functions with finitely many branches, and each branch has finitely many turning points. If $u/t = k(y)$, then

$$|K(y) - b| \leq \frac{B_5 R^2}{rG(x)} + \frac{B_6}{3\mu r N^2} \leq \frac{B_7 R^2}{rN}, \quad (6.3)$$

for some constant B_7 . For

$$r > 4B_7 R^2 / N, \quad (6.4)$$

the right-hand side of (6.3) is less than $\frac{1}{4}$, and $a = [[y]]$, $b = [[z]]$. Thus the point (a, b) in the y, z plane is an integer point lying close to an arc of the curve $z = K(y)$. The relation between the functions $g(x)$ and $K(y)$ is one that is usually encountered when evaluating a saddle point integral in the Poisson summation formula. We have

$$K'(y) = k(y), \quad k'(y) = -1/g''(k(y)).$$

We see that $K(y) = g(x) - xg'(x) - \beta$, and that $K(y)$ changes by

$$\int K'(y) dy = - \int xg''(x) dx. \quad (6.5)$$

Using the bound (4.12) for $g''(x)$ and the definition (3.11) of $G(x)$, we have

$$\begin{aligned} \int x g''(x) dx &\ll \int \frac{NR^2}{r^3} \left(\frac{r}{rx+s} \right)^3 \frac{R^2}{N^2} x dx \\ &\ll \frac{R^2}{rN^2} \int |G'(x)| dx \ll \frac{R^2}{rN^2} \max G(x). \end{aligned}$$

We impose a condition

$$G(x) \leq rN^2/B_8 R^2 \quad (6.6)$$

with a suitable constant B_8 , to ensure that $K(y)$ changes by less than $\frac{1}{2}$, and that only one integer b can be within $\frac{1}{4}$ of a value of $K(y)$.

We note that $g''(x)$ changes sign at most once, and that on an interval on which $g''(x)$ has constant sign, then y changes by

$$\int dy = \int \frac{dy}{dx} dx = - \int g''(x) dx. \quad (6.7)$$

Consider an interval J on which (6.1) holds and $g''(x)$ has constant sign and $\int x g''(x) dx$ changes by less than $\frac{1}{2}$. Comparing the integrals in (6.5) and (6.7) taken over J , we see that y changes on J by at most

$$\begin{aligned} 2B_7 \frac{R^2}{rN} \max_J \frac{1}{x} &\ll \frac{R^2}{rN} \max \frac{t}{u} \ll \frac{R^2}{rN} \max t \\ &\ll \frac{R^2}{rN} \frac{rQ}{NR^2} \max G\left(\frac{u}{t}\right) \ll \frac{Q}{N^2} \frac{N^2}{B_3 Q} < \frac{1}{2}, \end{aligned}$$

if $G(x)$ satisfies (5.10) and B_3 is sufficiently large. Hence a takes only one value in (6.2) when b is determined.

Without the condition (6.6), there would be several possible values of b , each corresponding to a unique integer a , with the conditions (6.2) and (6.3) holding. This leads to the problem of integer points close to a curve considered in [5].

We consider an interval I on which (6.1) and (6.2) hold with particular integers a and b . Suppose that the rationals u/t in I correspond to L consecutive minor arcs. We want to gain information when L is large. We number the minor arcs J_d in order, and pick points $x_i = u_i/t_i$ in I corresponding to minor arcs $J_{d(i)}$, with $i = 1, \dots, 8$ for which $d(i+1) - d(i) \geq \frac{1}{16}L$, so that $G(x_{i+1}) - G(x_i) \gg LN$. By Cauchy's mean value theorem, there are points $y_1, \dots, y_4$ with $x_{2i-1} < y_i < x_{2i}$ for which

$$\frac{\alpha - a - g'(y_i)}{G'(y_i)} \ll \frac{1}{LN} \frac{R^2}{rG(y_i)},$$

and thus

$$\alpha - a - g'(y_i) \ll \frac{R^2}{LN(ry_i + s)},$$

a sharpening of (6.2) by a factor $1/L$.

We repeat the process to eliminate $\alpha - a$. There are points z_1, z_2 with $y_{2i-1} < z_i < y_{2i}$ for which

$$\frac{g''(z_i)}{G'(z_i)} \ll \frac{1}{L^2 N^2} \frac{R^2}{rz_i + s},$$

and thus

$$g''(z_i) \ll \frac{1}{L^2} \frac{R^4}{N(rz_i + s)^3}. \quad (6.8)$$

From the formula (4.10) for $g''(x)$ we see that

$$\left(\frac{s_1/r_1 + z_i}{s/r + z_i}\right)^3 - \frac{\mu r^3}{\mu_1 r_1^3} \ll \frac{(rz_i + s)^3}{NR^2} \frac{1}{L^2} \frac{R^4}{N(rz_i + s)^3} \ll \frac{1}{L^2} \frac{R^2}{N^2},$$

which implies that

$$\frac{\mu_1(r_1 x + s_1)^3}{\mu(rx + s)^3} - 1 \ll \frac{1}{L^2} \frac{R^2}{N^2} \quad (6.9)$$

for $x = z_1$ and $x = z_2$. The left-hand side of (6.9) is monotone, since

$$\frac{r_1 x + s_1}{rx + s} = \frac{r_1}{r} + \frac{C}{r(rx + s)},$$

where C is an entry in the magic matrix M of § 3. Thus (6.9) holds for $z_1 \leq x \leq z_2$, an interval corresponding to at least $\frac{1}{32}L$ minor arcs. Giving x a rational value u/t in this interval, we have

$$\frac{\mu_1(r_1 u + s_1 t)^3}{\mu(ru + st)^3} - 1 \ll \frac{1}{L^2} \frac{R^2}{N^2},$$

and so the Third Condition (3.3) holds at u/t in the stronger form

$$\frac{\mu_1(t, u)(r_1 u + s_1 t)^3}{\mu(t, u)(ru + st)^3} - 1 \ll \frac{1}{L^2} \frac{R^2}{N^2} + \frac{n}{M} \ll \frac{1}{L^2} \Delta_2, \quad (6.10)$$

where we have used (5.3).

We obtain a corresponding strengthening in the First Condition (3.1) by observing that

$$\begin{aligned} \frac{r_1 z_2 + s_1}{r z_2 + s} - \frac{r_1 z_1 + s_1}{r z_1 + s} &= \frac{C}{r(r z_2 + s)} - \frac{C}{r(r z_1 + s)} \\ &= 3\mu C(G(z_2) - G(z_1)) \gg CL/R^2. \end{aligned}$$

Thus (6.9) gives

$$C \ll \frac{1}{L^3} \frac{R^4}{N^2} \ll \frac{1}{L^3} \Delta_1 Q^2, \quad (6.11)$$

improving the First Condition (3.1), which can be written as $C/q q_1 \leq \Delta_1$.

7. Counting coincident pairs of minor arcs

The numbers Δ_1 to Δ_4 in the Coincidence Conditions (3.1) to (3.4) depend on the size of q . In Step 4 of the method we subdivide into ranges $Q \leq q < 2Q$. We

apply the Large Sieve (Lemma 3.6 of [9]) separately to each range. We pick a positive integer V depending on Q . The structure of the large sieve allows us to replace the number of coincidences measured by $\Delta_1, \dots, \Delta_4$ by V times the number of coincidences measured by $\Delta_1/V, \Delta_2, \Delta_3$ and Δ_4 in the Coincidence Conditions (3.1) to (3.4). This restricts magic matrices to those whose lower left entry satisfies

$$C \leq \Delta_1 q q_1 / V \ll \Delta_1 Q^2 / V \ll R^4 / N^2 V. \quad (7.1)$$

We suppose for simplicity that

$$f''(x) \asymp M / NR^2 \asymp T / M^2$$

for the longer range $(1 - \delta)M \leq x \leq (2 + \delta)M$, for some $\delta > 0$. The numerators a of rational approximations a/q to $\frac{1}{2}f''(x)$ lie in some range $P \leq a \leq B_9 P$ with $P/Q \asymp M / NR^2$. If necessary we can reduce B_9 closer to unity by a finite subdivision of the range for P before applying the large sieve.

We consider the action

$$\begin{pmatrix} a_1 \\ q_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} a \\ q \end{pmatrix}$$

of the magic matrix. We have

$$Ca = (\theta - D)q, \quad Bq = (\varphi - A)a, \quad (7.2)$$

where $\theta = q_1/q$ and $\varphi = a_1/a$ are bounded away from zero and infinity. Eliminating a/q gives

$$\theta A + \varphi D = 1 + \theta \varphi. \quad (7.3)$$

Thus

$$B \ll (1 + |D|) \frac{P}{D} \ll \frac{P}{Q} + \frac{P^2}{Q^2} |C|. \quad (7.4)$$

We note that if $C \ll Q/P$ then $\theta - D$ is bounded, so that A and D are bounded, and either $BC = 0$, or B and C are also bounded, and similarly if $B \ll P/Q$. We classify magic matrices as follows:

- (1) the identity matrix and a bounded number of other matrices with A and D bounded, B and C both non-zero;
- (2) triangular matrices with $B = 0$ or $C = 0$;
- (3) non-triangular matrices with $B \gg P/Q$ and $C \gg Q/P$.

Type 3 matrices are usually the most important, but if P/Q is very large or very small, then they may not occur. For Type 3 matrices we have a symmetric form of (7.4):

$$|B| \asymp |C| P^2 / Q^2. \quad (7.5)$$

We assume that the conditions of Lemma 2.4 of [9] hold. In the proof of this lemma, Watt and the author showed that for a Type 3 matrix, the Third Condition (3.3) can hold only for a short interval of x . If $P \gg Q$, then $\frac{1}{2}f''(x)$ lies in a short interval of length $\ll \Delta_2 / |C|$. If $Q \gg P$, then $2/f''(x)$ lies in an interval of length $\ll \Delta_2 / |B|$. By (7.5) this also implies that $\frac{1}{2}f''(x)$ lies in an interval of length $\ll \Delta_2 / |C|$.

For simple sums we treat matrices of Types 1 and 2 trivially, in that the matrix may give a coincidence involving any minor arc. We note that there are a bounded number of matrices of Type 1, and that for Type 2 matrices

$$B \ll \Delta_2 P/Q, \quad C \ll \Delta_2 Q/P,$$

so that triangular matrices only arise when P/Q is very small or very large.

Now consider the magic matrix to be fixed. Our aim is to show that minor arcs on which all four Coincidence Conditions hold with this particular magic matrix are widely separated. Let U be an integer to be chosen below. We take the Farey sequence of order $[2R/\sqrt{U}]$ and refine it to a sequence of rationals about U/R^2 apart. If e/r and f/s are consecutive Farey fractions with $|f/s - e/r| \leq U/R^2$, then we leave the interval $[e/r, f/s]$ alone, but if $U/R^2 < |f/s - e/r| = 1/rs$, then we fill the gap. Suppose that $r > s$. We insert fractions

$$\frac{e_i}{r_i} = \frac{e + ft_i}{r + st_i} = \frac{f}{s} \pm \frac{1}{s(r + st_i)},$$

which are adjacent to f/s in their respective Farey sequences. We choose t_i by $t_0 = 0$, and for $i \geq 1$,

$$r + st_i = 2R^2/isU + O(1),$$

the sequence stopping at t_i when $1/s(r + st_i) \leq U/R^2$. We note that since $2r - 1 \geq r + s > 2R/\sqrt{U}$, we have $r + st_i \geq r \geq R/\sqrt{U}$. In our enlarged Farey sequence the gaps between consecutive fractions have lengths between $U/4R^2$ and $4U/R^2$. We call this sequence the *reference fractions*. This construction is another way of achieving uniform Diophantine approximation.

We apply the construction of § 3 to the rational values of $\frac{1}{2}f''(x)$ lying in an interval between two consecutive reference fractions. The reference fraction with larger denominator becomes e/r for the construction. If e/r lies in the Farey sequence of order $[2R/\sqrt{U}]$, then the reference fraction at the other end of the interval becomes f/s . If e/r is constructed as e_i/r_i with $i \geq 1$ as above, then the fraction f/s used to construct e_i/r_i becomes the f/s for § 3. The fractions e/r and f/s chosen may not correspond to values of $\frac{1}{2}f''(x)$ in the interval $M \leq x \leq 2M$, but they must correspond to values of $\frac{1}{2}f''(x)$ in the interval $(1 - \delta)M \leq x \leq (2 + \delta)M$, provided that M is sufficiently large in terms of δ . The interval between two consecutive reference fractions corresponds to about U arcs in order of magnitude, necessarily minor arcs. For some constant $B_{10} \geq 1$ there are between U/B_{10} and $B_{10}U$ minor arcs between the minor arcs containing two consecutive reference fractions.

The minor arcs for which the Coincidence Conditions hold with our fixed magic matrix fall into blocks of consecutive minor arcs. Let l be the length of such a block. If the l minor arcs contain two or more reference fractions, then $l \geq B_{10}U$. If the l minor arcs contain at most one reference fraction, then there is an interval between two consecutive reference fractions containing at least $\frac{1}{2}l$ minor arcs of the block, possibly including the minor arc containing the reference fraction. We may suppose that there is an interval between two consecutive reference fractions containing L consecutive minor arcs on which the Coincidence Conditions hold, with L a power of 2 satisfying $L \geq \min(\frac{1}{4}l, U/2B_{10})$. The choice of U can depend on L , in which case, if several blocks occur corresponding to different values of

L , then we repeat the argument with different choices of U and different systems of reference fractions. For simplicity we will take U fixed.

The conditions on U come from those on $G(x)$. We note that $G(x) \leq (B_{10}U + O(1))N$. To satisfy (5.3) we must have

$$U \leq (MR^2)^{1/3}/B_{11}N, \quad (7.6)$$

for some constant $B_{11} > 0$. To satisfy (5.10) we need

$$U \leq N/B_{12}Q, \quad (7.7)$$

for some constant $B_{12} > 0$. To satisfy (6.6) we must have

$$U \leq rN/B_{13}R^2,$$

for some constant $B_{13} > 0$. Since $r \geq R/\sqrt{U}$, this condition is ensured when

$$U \leq \frac{1}{B_{14}} \left(\frac{N}{R} \right)^{2/3}, \quad (7.8)$$

for some constant $B_{14} > 0$. We take U to be the largest integer satisfying

$$U \leq \frac{1}{B_{15}} \left(\frac{N}{Q} \right)^{2/3} \asymp \Delta_4^{-2/3}, \quad (7.9)$$

where B_{15} is chosen so that (7.7) and (7.8) hold. The condition (7.6) also holds, provided that the parameters of the construction satisfy

$$B_{16}N^5 \leq MR^4, \quad (7.10)$$

for a suitable constant B_{16} .

First we consider triangular matrices. If any interval between two consecutive reference fractions contains a block of L consecutive minor arcs on which the Coincidence Conditions hold, then by (6.10), the Third Condition (3.3) holds with Δ_2 replaced by $\ll \Delta_2/L^2$. We deduce that

$$B \ll \frac{1}{L^2} \Delta_2 \frac{P}{Q}, \quad C \ll \frac{1}{L^2} \Delta_2 \frac{Q}{P}$$

for the upper and lower triangular matrices respectively. Thus summing L through powers of 2, we see that lower triangular matrices contribute

$$\ll \sum_L \sum_{C \ll \Delta_2 Q/L^2 P} \frac{LM}{NU} \ll \frac{1}{U} \Delta_2 \frac{MQ}{NP} \ll \frac{R^4}{N^2 U}$$

coincident pairs of minor arcs. Similarly, upper triangular matrices give $\ll M^2/N^4 U$ coincident pairs. Thus triangular matrices contribute

$$\ll \frac{1}{U} \left(\frac{R^4}{N^2} + \frac{M^2}{N^4} \right) \ll \frac{R^2}{Q^2} \frac{\Delta_2}{U} (P^2 + Q^2) \quad (7.11)$$

pairs of coincident minor arcs with $Q \leq q < 2Q$.

We save less with greater effort on the Type 3 matrices. Suppose that some interval between two consecutive reference fractions contains a block of L consecutive minor arcs on which the four Coincidence Conditions hold. Combin-

ing (6.11) and (7.1) we have

$$C \ll \frac{R^4}{N^2} \min\left(\frac{1}{L^3}, \frac{1}{V}\right). \quad (7.12)$$

By (6.10) the Third Condition (3.3) holds with Δ_2 replaced by some smaller number $\ll \Delta_2/L^2$. From the proof of Lemma 2.4 of [9], we see that the fraction $(eu + ft)/(ru + st)$ must lie in an interval of length $\ll \Delta_2/L^2 |C|$ containing $U_0 \ll \Delta_2 R^2/L^2 |C|$ minor arcs.

For $P \gg Q$ there are $\ll |C| P^2/Q^2$ possible Type 3 matrices with lower left entry C . We sum over C satisfying (7.12) to estimate the contribution of blocks of $\gg L$ consecutive minor arcs with $Q \leq q < 2Q$ as

$$\begin{aligned} &\ll \sum_C L |C| \frac{P^2}{Q^2} \left(1 + \frac{U_0}{U}\right) \\ &\ll \frac{P^2 R^8}{Q^2 N^4} \left(\min\left(\frac{1}{L^5}, \frac{L}{V^2}\right) + \frac{1}{U} \min\left(\frac{1}{L^4}, \frac{1}{LV}\right)\right). \end{aligned}$$

When we sum L through powers of 2, then this bound becomes

$$\ll \frac{P^2 R^8}{Q^2 N^4} \left(\frac{1}{V^{5/3}} + \frac{1}{UV}\right). \quad (7.13)$$

Similarly for $P \ll Q$ there are $\ll |B| P^2/Q^2$ possible Type 3 matrices. Summing over B and using (7.5), we obtain the bound (7.13) in this case also.

The form of the bound for Type 3 matrices leads to the choice

$$V \asymp U^{3/2} \asymp 1/\Delta_4. \quad (7.14)$$

The total number of coincidences between minor arcs is

$$\begin{aligned} &\ll \frac{R^2}{Q^2} V \left(PQ + \Delta_2 \frac{1}{U} (P^2 + Q^2) + \Delta_1 \Delta_2 \frac{P^2 Q^2}{V^{5/3}} \right) \\ &\ll \frac{R^2}{Q^2} \left(\frac{PQ}{\Delta_4} + \Delta_2 \frac{(P^2 + Q^2)}{\Delta_4^{1/3}} + \Delta_1 \Delta_2 \Delta_4^{2/3} P^2 Q^2 \right) \\ &\ll \left(\frac{Q}{R} \right)^{2/3} \left(\frac{M}{R} + \frac{M^2}{N^{11/3} R^{1/3}} + \frac{R^{11/3}}{N^{5/3}} + \frac{M^2 R^{14/3}}{N^{20/3}} \right). \end{aligned} \quad (7.15)$$

In (7.15) we save a factor $\Delta_4^{2/3}$ on Type 3 matrices, but we lose powers of Δ_4 on Type 1 and Type 2 matrices. Consequently we must choose N , the length of the short sums, rather smaller than in previous papers of this series.

We note that if $V > 4\Delta_1 Q^2$, then all magic matrices are upper triangular, and the last term in (7.15) may be omitted. For $P \geq Q$ with

$$V \asymp \Delta_1 Q^2 \asymp R^4/N^2, \quad (7.16)$$

the total number of coincidences between minor arcs with $Q \leq q < 2Q$ is

$$\ll \frac{R^2}{Q^2} (\Delta_1 PQ^3 + \Delta_1 \Delta_2 \Delta_4^{2/3} P^2 Q^2) \ll \left(\frac{Q}{R} \right)^{2/3} \left(\frac{MR^4}{N^3} + \frac{M^2 R^{14/3}}{N^{20/3}} \right). \quad (7.17)$$

Similarly, if $P \leq Q/B_{17}$ for some sufficiently large constant B_{17} , then by (7.4) all matrices are lower triangular for $V > B_{18} \Delta_1 P^2$ with some sufficiently large

constant B_{18} . For some choice of V with

$$V \asymp \Delta_1 P^2 \asymp M^2 R^2 / N^4, \quad (7.18)$$

the total number of coincidences between minor arcs with $Q \leq q < 2q$ is

$$\ll \frac{R^2}{Q^2} (\Delta_1 P^3 Q + \Delta_1 \Delta_2 \Delta_4^{2/3} P^2 Q^2) \ll \left(\frac{Q}{R}\right)^{2/3} \left(\frac{M^3}{N^5} + \frac{M^2 R^{14/3}}{N^{20/3}}\right). \quad (7.19)$$

8. The exponential sum result

We can now give a new bound for the exponential sum (1.1). Further difficulties arise when we consider a family of exponential sums as in [10]. We treat these later.

THEOREM 1. *Let $F(x)$ be a function four times differentiable on the interval $1 - \delta \leq x \leq 2 + \delta$ for some $\delta > 0$, whose derivatives satisfy the following conditions:*

$$|F^{(r)}(x)| \leq C_1 \quad (8.1)$$

for $r = 3, 4$;

$$|F^{(r)}(x)| \geq 1/C_1 \quad (8.2)$$

for $r = 3$, where C_1 is a positive constant. Let

$$S = \sum_{m=M}^{M_2} e\left(TF\left(\frac{m}{M}\right)\right),$$

where M and M_2 are positive integers, T is a large real number and $M \leq M_2 < 2M < T$. Suppose that for some $C_2 \geq 1$ either Case 1 or Case 2 holds:

Case 1: $M \leq C_2 \sqrt{T}$ and (8.2) holds for $r = 4$ also;

Case 2: $M \geq C_2^{-1} \sqrt{T}$ and (8.1) and (8.2) hold for $r = 2$ also and

$$|F''(x)F^{(4)}(x) - 3(F^{(3)}(x))^2| \geq C_3 \quad (8.3)$$

for some positive constant C_3 .

If M is sufficiently large (in terms of C_1 and δ), then for any $\varepsilon > 0$ we have

$$S \ll \begin{cases} M^{1/2} T^{89/570 + \varepsilon} & \text{for } T^{49/114} \leq M \leq T^{65/114}, \\ M^{7/20} T^{13/60 + \varepsilon} + M^{13/20} T^{11/120 + \varepsilon} & \text{for } M \leq T^{49/114}, \\ M^{7/20} T^{29/120 + \varepsilon} + M^{13/20} T^{1/15 + \varepsilon} & \text{for } M \geq T^{65/114}. \end{cases}$$

The implied constants depend on C_1 , C_2 , C_3 , δ and ε .

Proof. First we note a weakening in the conditions of Theorem 1 compared with the corresponding Theorem 1 in [9]. In Case 1 we have dropped the conditions on F'' . By (8.1) for $F^{(3)}$ we see that F'' lies in a bounded interval, and we replace $F(x)$ by $F(x) + aM^2 x^2/T$ for some suitable integer a , chosen so that

$$1 \leq F''(x) + 2aM^2/T \leq 1 + (1 + 2\delta)C_1 + 4C_2^2.$$

Then (8.1) and (8.2) hold with a larger constant in place of C_1 . We can thus assume that (8.1) and (8.2) hold for $r = 2$ in all cases, which gives the assumption $f''(x) \asymp T/M^2$ in § 7. The conditions (8.1), (8.2) and (8.3) for the appropriate

values of r are now those of Theorem 1 in [9] and they ensure that Lemma 2.4 of that paper is valid.

We recall some details of the construction in [9]. The parameters N and R are chosen at the outset, with

$$N \ll MT^{-1/4}, \quad (8.4)$$

$$R^2 NT \geq 2C_1 M^3 > (R-1)^2 NT, \quad (8.5)$$

$$R \leq N \leq R^2, \quad (8.6)$$

$$NR \ll M. \quad (8.7)$$

The condition (8.7) can be removed by modifying Lemma 2.1 of [9]. However the choices of N and R in this paper will all satisfy (8.7).

On minor arcs the rational number a/q chosen as a value of $\frac{1}{2}f''(x)$ has $q \geq R$. By Lemma 2.1 of [9], for Q any power of 2 with $Q \geq R$, there are $\ll MR/NQ$ minor arcs for which q is chosen to be at least Q . If the rational number chosen has $Q \leq q < 2Q$, then there are $\gg Q^2/R^2$ rational values of $\frac{1}{2}f''(x)$ on the minor arc with denominators in the range Q to $16Q$. This is shown in § 9 of [6].

The bound used in [8] for the sum S of (1.1) is

$$|S|^5 \ll \frac{M^5 \log^5 N}{N^{5/2}} + \sum_Q \left(\frac{MR}{NQ} \right)^4 \frac{R^5}{Q^{5/2}} \left(\frac{R^2}{Q^2} BH^{11+\varepsilon} \left(1 + \frac{Q}{N} \right) \right)^{1/2} \log^9 N,$$

where Q is summed through powers of 2 with $Q \geq \frac{1}{2}R$, and H depends on Q with $H \asymp NQ/R^2$. Here B denotes the number of coincident pairs of minor arcs with $Q \leq q < 2Q$. We simplify this inequality to

$$|S|^5 \ll \frac{M^5 T^\varepsilon}{N^{5/2}} + \frac{M^4 N^{3/2} T^\varepsilon}{R^3} \sum_Q \frac{R^2}{Q^2} B^{1/2}.$$

For $Q \geq N/16B_{12}$ we estimate B trivially as the square of MR/NQ in order of magnitude, which is an upper bound for the number of minor arcs on which a/q is chosen with $q \geq Q$. For $Q \leq N/16B_{12}$ we use the construction of §§ 3 to 7; here B_{12} is the constant in (7.7). We can choose the parameters so that (7.15), (7.17) or (7.19) gives

$$B \ll \left(\frac{Q}{R} \right)^{2/3} \frac{M^2 R^{14/3}}{N^{20/3}}. \quad (8.8)$$

The sum over Q converges, and

$$|S|^5 \ll M^5 T^\varepsilon / N^{11/6} R^{2/3},$$

as promised in the introduction.

We have to choose N (which determines R by (8.5)) so that (8.6), (8.7) and (7.10) hold; the condition (7.10) implies (8.4). For $T^{49/114} \ll M \ll T^{65/114}$ we choose N and R by

$$N \asymp MT^{-17/57}, \quad R \asymp MT^{-20/57},$$

and V by (7.14) to equalise the first and fourth terms in (7.15). For $T^{1/3} \ll M \ll T^{49/114}$ we choose V by (7.16). For $M \gg T^{5/12}$ we choose N and R to equalise the two terms in (7.17) by

$$N \asymp M^{1/2} T^{-1/12}, \quad R \asymp M^{5/4} T^{-11/24}.$$

For smaller M this choice would give $V < 1$, so we take $N \asymp R^2 \asymp (M^3/T)^{1/2}$, with $V \ll 1$. The bound of the theorem is trivial for $M \ll T^{1/3}$. For $T^{65/114} \ll M \ll T^{2/3}$ we choose V by (7.18). For $M \ll T^{7/12}$ we choose N and R to equalise the two terms in (7.19) by

$$N \asymp M^{3/2} T^{-7/12}, \quad R \asymp M^{3/4} T^{-5/24}.$$

For larger M this choice would give $V < 1$, so we take

$$N \asymp M^{1/2}, \quad R \asymp M^{5/4} T^{-1/2}$$

with $V \ll 1$. For $M \gg T^{2/3}$ we get a better result by treating all arcs as major, as remarked in [9]; this corresponds to the 'exponent pair $(\frac{1}{6}, \frac{2}{3})$ ' in the classical theory [3]. Thus Theorem 1 remains true for $M \ll T^{1/3}$ and for $M \gg T^{2/3}$. The conditions (8.6), (8.7) and (7.10) hold in all cases. We find that (8.7) and the first inequality of (8.6) are sharp when M is close to $T^{2/3}$, and that the second inequality of (8.6) is sharp for $T^{1/3} \ll M \ll T^{5/12}$.

9. Extending the Taylor series

If $f(x)$ is five times differentiable, with the natural bounds

$$f^{(5)}(x) \ll |f^{(3)}(x)|/M^2 \ll 1/M^2 NR^2,$$

then we take the Taylor polynomials about m in § 3 to one degree higher, with an explicit term in $v = \frac{1}{24} f^{(4)}(m)$. We pick a system of reference fractions with parameter U , so that the variable n in the Taylor series satisfies $|n| \leq B_{10} NU$. The condition (5.3), $n^3 \ll MR^2$, allowed us to neglect terms in the fourth derivative. We replace (5.3) with the bound

$$n^2 \ll N^2 U^2 \ll MR, \quad (9.1)$$

which implies

$$n^4 f^{(5)}(x) \ll 1/N \quad (9.2)$$

in the Taylor series.

The mean value theorem for the third derivative gives

$$f^{(3)}(m+n) = f^{(3)}(m) + n f^{(4)}(m) + O(n^2 \max |f^{(5)}(x)|),$$

so that

$$6\mu(t, u) = 6\mu + 24nv + O(n^2/M^2 NR^2).$$

To eliminate $\mu(t, u)$ from the Third Condition, we write

$$\begin{aligned} \frac{1}{\mu(t, u)(ru + st)^3} &= \frac{1}{\mu(ru + st)^3} \left(1 + \frac{4nv}{\mu} + O\left(\frac{n^2}{M^2}\right) \right)^{-1} \\ &= \frac{1}{\mu(ru + st)^3} \left(1 - \frac{4nv}{\mu} + O\left(\frac{n^2}{M^2}\right) \right) \\ &= \frac{1}{\mu(ru + st)^3} - \frac{4vt}{3\mu^3 r(ru + st)^4} + O\left(\frac{NR^2}{MQ^3} \left(1 + \frac{n^2}{M}\right)\right), \end{aligned} \quad (9.3)$$

where we have used (3.11). The Third Condition can be expressed as

$$\frac{1}{\mu(t, u)(ru + st)^3} - \frac{1}{\mu_1(t, u)(r_1u + s_1t)^3} \ll \frac{R^2 NR^2}{N^2 Q^3}, \quad (9.4)$$

and the error term in (9.4) dominates the error term in (9.3), by (9.1) and (9.2). We substitute (9.3) and the corresponding relation for $\mu_1(t, u)$ to make the left-hand side of (9.4) into

$$\frac{3}{2t^3} \left(g''\left(\frac{u}{t}\right) - h''\left(\frac{u}{t}\right) \right) + O\left(\frac{NR^2}{Q^3} \left(\frac{1}{M} + \frac{n^2}{M^2}\right)\right),$$

where $h(x)$ is defined by

$$h(x) = \frac{4v}{27\mu^3 r^3 (rx + s)^2} - \frac{4v_1}{27\mu_1^3 r_1^3 (r_1x + s_1)^2}. \quad (9.5)$$

Hence the Third Condition implies that

$$g''\left(\frac{u}{t}\right) - h''\left(\frac{u}{t}\right) \ll \frac{R^2 t^3 NR^2}{N^2 Q^3} \asymp \frac{R^4}{N(ru/t + s)^3}. \quad (9.6)$$

Next we examine the mean value theorem for the second derivative:

$$f''(m + n) = f''(m) + nf^{(3)}(m) + \frac{1}{2}n^2 f^{(4)}(m) + O(n^3 \max |f^{(5)}(x)|),$$

which gives

$$\frac{2(eu + ft)}{ru + st} + 2\lambda(t, u) = \frac{2e}{r} + 2\lambda + 6n\mu + 12n^2v + O\left(\frac{n^3}{M^2 NR^2}\right),$$

and

$$\begin{aligned} n &= \frac{t}{3\mu r(ru + st)} - \frac{2n^2v}{\mu} + O\left(1 + \frac{n^3}{M^2}\right) \\ &= G\left(\frac{u}{t}\right) - \frac{2vt^2}{9\mu^3 r^2 (ru + st)^2} + O\left(1 + \frac{n^3}{M^2}\right). \end{aligned} \quad (9.7)$$

Thus

$$\frac{n}{ru + st} - \frac{n_1}{r_1u + s_1t} = -\frac{1}{t} g'\left(\frac{u}{t}\right) + \frac{3}{4t} h'\left(\frac{u}{t}\right) + O(\Delta_3), \quad (9.8)$$

where we have used (9.2) to estimate the error term.

Finally, we expand the first derivative by the mean value theorem:

$$f'(m + n) = f'(m) + nf''(m) + \frac{1}{2}n^2 f^{(3)}(m) + \frac{1}{6}n^3 f^{(4)}(m) + O(n^4 \max |f^{(5)}(x)|),$$

which gives

$$\frac{c(t, u) + \delta(t, u)}{ru + st} = \frac{c + \delta}{r} + \frac{2en}{r} + 2n\lambda + 3n^2\mu + 4n^3v + O\left(\frac{1}{N}\right)$$

by (9.3). From the definition (3.13) of γ we have

$$\begin{aligned}\frac{\gamma}{ru+st} &= 2n\lambda + 3n^2\mu + 4n^3\nu - \frac{nt}{r(ru+st)} + O\left(\frac{1}{N}\right) \\ &= G(2\lambda + 3\mu(n-G) + 4n^2\nu) + O(1/N).\end{aligned}$$

We use (3.11) repeatedly to get

$$\gamma/(ru+st) = G(2\lambda + 3\mu(n-G) + 4G^2\nu) + O(1/N).$$

Thus

$$\gamma = \frac{t}{r} \left(n - G\left(\frac{u}{t}\right) + \frac{2\lambda}{3\mu} \right) + \frac{4\nu t^3}{27\mu^3 r^3 (ru+st)^2} + O(\Delta_4). \quad (9.9)$$

We can also substitute from (9.7) to obtain

$$\gamma = \frac{2\lambda t}{3\mu r} - \frac{2\nu t^3}{27\mu^3 r^3 (ru+st)^2} + O(\Delta_4).$$

Continuing as in § 5, we have

$$\frac{\gamma r}{t(ru+st)} - \frac{\gamma_1 r_1}{t(r_1 u + s_1 t)} = \frac{1}{4t} h'\left(\frac{u}{t}\right) + O(\Delta_3). \quad (9.10)$$

We are now ready to consider the Fourth Condition. From (9.9),

$$\gamma - \gamma_1 = \frac{nt}{r} - \frac{n_1 t}{r_1} - t g\left(\frac{u}{t}\right) + \frac{2\lambda t}{3\mu r} - \frac{2\lambda_1 t}{3\mu_1 r_1} + t h\left(\frac{u}{t}\right) + O(\Delta_4).$$

To linearise as in § 5 we require

$$ty^2(g''(x) - h''(x)) \ll \Delta_4, \quad (9.11)$$

where x is a value corresponding to the minor arc considered, and

$$y \ll (r^2 x^2 + s^2)/R^2.$$

The estimate (9.6) has the same strength as (4.12), so (9.11) follows from the Third Condition. We have a linear approximation of the same form as in § 5 with

$$\begin{aligned}\alpha' &= \alpha - g'(x_0) + h'(x_0), \\ \beta' &= \beta - g(x_0) + h(x_0) + x_0 g'(x_0) - x_0 h'(x_0).\end{aligned}$$

If (5.9) and (5.10) hold, then we recover (5.11) and (5.12). We can still divide by t in the Fourth Condition to get

$$|(\alpha - a)x + \beta - b - g(x) + h(x)| \leq B_5 \frac{R^2}{rG(x)}. \quad (9.12)$$

We turn to the Second Condition in the form (3.17). When the Fourth Condition holds, then the terms involving η and η_1 are $O(\Delta_3)$, and the term in $\bar{u}$ is given by (5.14). The left-hand side of (3.17) is

$$\begin{aligned}\left\| \frac{n}{ru+st} - \frac{n_1}{r_1 u + s_1 t} + \frac{\alpha - a}{t} + \frac{\gamma r}{t(ru+st)} - \frac{\gamma_1 r_1}{t(r_1 u + s_1 t)} \right\| + O(\Delta_3) \\ = \left\| -\frac{1}{t} g'\left(\frac{u}{t}\right) + \frac{1}{t} h'\left(\frac{u}{t}\right) + \frac{\alpha - a}{t} \right\| + O(\Delta_3), \quad (9.13)\end{aligned}$$

by (9.8) and (9.10). As in § 5, we have

$$\left| \frac{\alpha - a}{t} \right| < \frac{1}{8}, \quad \text{and} \quad \left| \frac{1}{t} g' \left(\frac{u}{t} \right) \right| < \frac{1}{8},$$

if B_1 and B_3 are sufficiently large. By (9.5) and (9.2) we have

$$\frac{1}{t} h' \left(\frac{u}{t} \right) \ll \left| \frac{vt^2}{\mu^3 r^2 (ru + st)^3} \right| \ll \frac{1}{MQ} G^2 \left(\frac{u}{t} \right) \ll \frac{1}{B_1},$$

and so if B_1 is sufficiently large, then the sum of the three terms in (9.13) is less than $\frac{1}{2}$, and we deduce that

$$|\alpha - a - g'(u/t) + h'(u/t)| \leq B_6 r G(x) / N^2 \quad (9.14)$$

as in § 5.

We continue as in § 6, defining $k(y)$ implicitly by

$$g'(k(y)) - h'(k(y)) = \alpha - y,$$

with

$$z = K(y) = (y - \alpha)k(y) + g(k(y)) - h(k(y)) - \beta;$$

we replace $g(x)$ by $g(x) - h(x)$ in all the formulae. For some constant B_{19} , the condition

$$r > B_{19} R^2 / N \quad (9.15)$$

ensures that, if $u/t = k(y)$, then

$$a = [[y]], \quad b = [[z]].$$

In place of (6.5) and (6.7) we have

$$\int K'(y) dy = - \int x(g''(x) - h''(x)) dx$$

$$\int dy = - \int (g''(x) - h''(x)) dx.$$

Since $g(x)$ and $h(x)$ are rational functions of bounded degree, there are boundedly-many subintervals of x on which the Third Condition holds in the form (9.6). As (9.6) has the same strength as (4.12), we see that if (6.6) holds with a suitable constant B_8 , then, on each subinterval, $K(y)$ changes by less than $\frac{1}{2}$, and that just one integer b can be less than $\frac{1}{4}$ from some value of $K(y)$. As before, we see that the integer a takes only one value on each subinterval, and that if the Coincidence Conditions persist for L consecutive minor arcs of the subinterval with $L \geq 32$, then there are rational numbers $u_1/t_1, u_2/t_2$ with

$$G(u_2/t_2) - G(u_1/t_1) \asymp LN, \quad (9.16)$$

but

$$g''(x) - h''(x) \ll \frac{1}{L^2} \frac{R^4}{N(rx + s)^3}$$

for x equal to u_1/t_1 and u_2/t_2 . By (9.6) the Third Condition holds at u_1/t_1 and u_2/t_2 in the form

$$\begin{aligned} \frac{\mu_1(t, u)(r_1 u + s_1 t)^3}{\mu(t, u)(ru + st)^3} - 1 &\ll \frac{1}{L^2} \frac{R^2}{N^2} + \frac{1}{M} \left(1 + \frac{n^2}{M}\right) \\ &\ll \Delta_2/L^2, \end{aligned} \quad (9.17)$$

by (9.2).

From (9.16) we have

$$\frac{\mu(t_1, u_1)}{\mu(t_2, u_2)} - 1 \ll \frac{LN}{M},$$

and similarly for $\mu_1(t, u)$. Thus

$$\frac{r_1 u_2 + s_1 t_2}{ru_2 + st_2} - \frac{r_1 u_1 + s_1 t_1}{ru_1 + st_1} \ll \frac{LN}{M} + \frac{1}{L^2} \frac{R^2}{N^2},$$

and as in § 6 we see that

$$C \ll \frac{1}{L^3} \frac{R^4}{N^2} + \frac{NR^2}{M}, \quad (9.18)$$

a sharpening of the First Condition.

Thus we have replaced (5.3) by (9.1), provided that the extra differentiability conditions hold. When U is chosen by (7.9), then (9.1) holds if

$$N^{10} \ll M^3 R^7. \quad (9.19)$$

Since $R \leq N$, the condition (9.19) implies (4.8).

10. Counting coincident pairs with a parameter

As Jutila noticed [12], we compare Taylor approximations to $f(x)$ on two different short intervals. In general these intervals do not overlap, and the values of the function at intermediate points do not enter the argument. We may replace $F(x)$ by a family of functions $F_i(x)$. For non-triangular matrices the argument of Lemma 2.4 of [9] goes through unchanged, and coincidence between $F_i(x)$ and $F_j(x)$ given by a fixed magic matrix occurs only on a short interval of values of x . For triangular matrices we need the functions $F_i(x)$ and $F_j(x)$ to be distinct. A useful sufficient condition is that $F_i(x) = F(x, y_i)$, where $y_1, \dots, y_l$ are well-spaced points in $[0, 1]$, and certain combinations of partial derivatives are bounded away from zero; see [10] for more details.

Accordingly we consider the sums

$$S_i = \sum_{m=M_1(i)}^{M_2(i)} e\left(TF\left(\frac{m}{M}, y_i\right)\right), \quad (10.1)$$

where $M \leq M_1(i) \leq M_2(i) < 2M$, and $y_1, \dots, y_l$ lie in $[0, 1]$ with

$$y_{i+1} - y_i \geq 1/J \quad (10.2)$$

for $i \leq I - 1$. If necessary we subdivide the sums S_i so that we may assume that

$$\max_i M_2(i) - \min_i M_1(i) \leq M/B_{20}, \quad (10.3)$$

for any sufficiently large constant B_{20} .

We write ∂_1, ∂_2 for $\partial/\partial x, \partial/\partial y$, and F_1 for $\partial_1 F$, etc. We consider together all the minor arc vectors corresponding to different values of the parameter. To avoid complicating the notation further, we suppose that $2a/q$ is a value of $T\partial_1^2 F(x/M, z)$ and $2a_1/q_1$ is a value of $T\partial_1^2 F(x/M, z_1)$, where z and z_1 lie in the sequence $y_1, \dots, y_I$. When counting coincident vectors as in § 7, we may choose to fix z and z_1 , and then count rationals a/q (call this counting horizontally), or we may fix a/q and z , and then count well-spaced values of the parameter $y_i = z_1$ (call this counting vertically). The conditions imposed on the function $F(x, y)$ in [10] mean that x and y enter the definition of $F(x, y)$ independently. We always count horizontally in the First and Fourth Conditions, but we may count vertically in the Third Condition.

Type 1 matrices are the identity and a bounded number of other matrices with entries too small for the argument used on Type 3. For these we consider coincidences between the minor arc with rational point $(eu + ft)/(ru + st)$ and parameter z (fixed), and the minor arc with rational point $(e_1u + f_1t)/(r_1u + s_1t)$ (fixed by the matrix) and parameter z_1 (variable). We compare 'vertically' in the y variable. Let z_2 be another value of the parameter y for which there is a coincidence, and let $\mu_2(t, u)$ be the corresponding value of the third derivative. Then the Third Condition in the form (9.17) gives

$$\frac{\mu_i(t, u)(r_1u + s_1t)^3}{\mu(t, u)(ru + st)^3} - 1 \ll \frac{\Delta_2}{L^2}$$

for $i = 1$ and 2 , so that

$$\frac{\mu_2(t, u)}{\mu_1(t, u)} - 1 \ll \frac{\Delta_2}{L^2}. \quad (10.4)$$

In Lemma 2.1 of [10], Watt and the author showed that, under certain conditions which follow from the hypotheses of Theorem 2 of [10], the inequality (10.4) implies that $z_1 - z_2 \ll \Delta_2/L^2$. If $z_1 = y_j$ and $z_2 = y_k$, then by the spacing condition (10.2) we have $j - k \ll \Delta_2 J/L^2$.

We can now count the number of coincident pairs of minor arcs given by a Type 1 matrix. There are $\ll MR^2/NQ^2$ minor arcs with $Q \leq q < 2Q$. Suppose that the interval between two consecutive reference fractions contains a block of L consecutive minor arcs which coincide with the corresponding blocks both with parameter y_j and with parameter y_k . We allow $j = k$ here. If $j \neq k$, then $L^2 \ll \Delta_2 J/|j - k|$. Each interval between two consecutive reference fractions contains (at most) a bounded number of such blocks. Summing L through powers of 2 with $L \ll U$, we get

$$\ll \frac{IMR^2}{NQ^2} \sum_L \frac{L}{U} \left(1 + \Delta_2 \frac{J}{L^2}\right) \ll \frac{IMR^2}{NQ^2} \left(1 + \Delta_2 \frac{J}{U}\right)$$

coincident pairs of minor arcs with $Q \leq q < 2Q$ for each Type 1 matrix.

Next we consider triangular matrices. If any interval between two consecutive reference fractions contains a block of L consecutive minor arcs on which the

Coincidence Conditions hold, then by (9.17) the Third Condition (3.3) holds with $\ll \Delta_2$ replaced by $\ll \Delta_2/L^2$. If $B \ll \Delta_2 P/L^2 Q$, or if $C \ll \Delta_2 Q/L^2 P$, then for each pair of values of z and z_1 , we argue as in § 7.

If $z \neq z_1$, then a triangular matrix can have B or C larger than in § 7, with

$$\frac{\Delta_2 P}{L^2 Q} \ll |B| \leq \frac{P}{Q}, \quad \frac{\Delta_2 Q}{L^2 P} \ll |C| \leq \frac{Q}{P}. \quad (10.5)$$

In Lemma 2.2 of [10], Watt and the author showed, for any fixed z and z_1 , under certain conditions which follow from the hypotheses of Theorem 2 of [10], that $f''(x)$ must lie in an interval of length $\asymp \Delta_2/L^2|C|$, or $\asymp \Delta_2 P^2/L^2|B|Q^2$, containing rational points corresponding to $\ll U_0 \asymp \Delta_2 R^2/L^2|C|$ consecutive minor arcs, with a similar formula in the upper triangular case. Since L minor arcs fit into this interval, we must have

$$L^3 \ll \Delta_2 R^2/|C|. \quad (10.6)$$

This inequality also holds for Type 3 matrices, but does not lead to any sharper estimates there.

At this point the argument becomes analogous to the treatment of Type 3 matrices in § 7. For each pair of values z and z_1 of the parameter, the contribution of blocks of $\asymp L$ consecutive minor arcs with $Q \leq q < 2Q$ is

$$\ll \sum_C L \left(1 + \frac{U_0}{U}\right) \ll \sum_C L \left(1 + \frac{\Delta_2 R^2}{UL^2|C|}\right),$$

where L and C are constrained by (10.6). We sum L through powers of 2 satisfying (10.6) to get

$$\begin{aligned} &\ll \sum_C \left(\Delta_2 \frac{R^2}{|C|}\right)^{1/3} + \sum_C \Delta_2 \frac{R^2}{U|C|} \\ &\ll (\Delta_2 R^2 Q^2/P^2)^{1/3} + \Delta_2 \Delta_4^{2/3} R^2 \log N. \end{aligned}$$

We can omit the first term if

$$N^4 \ll MR^3(\log N)^{3/2}. \quad (10.7)$$

Similarly, the upper triangular matrices satisfying (10.5) give

$$\ll \left(\Delta_2 \frac{P^4 R^2}{Q^4}\right)^{1/3} + \Delta_2 \Delta_4^{2/3} \frac{P^2 R^2}{Q^2} \log N.$$

The first term can again be omitted if (10.7) holds.

Summing over z and z_1 , we have

$$\ll \frac{I^2 R^2}{Q^2} \Delta_2 \Delta_4^{2/3} (P^2 + Q^2) \log N + I^2 \left(\frac{R}{Q}\right)^{2/3} (\Delta_2 P Q)^{1/3} \left(\frac{P}{Q} + \frac{Q}{P}\right)$$

coincident pairs of minor arcs with $Q \leq q < 2Q$ given by Type 2 matrices. The corresponding estimate in [10] has a term in $1/PQ$, which we avoid by counting minor arcs rather than rational numbers. Instead we have an extra term arising from the maximum of L .

For Type 3 matrices (9.18) gives

$$C \ll \min\left(\frac{1}{V} \frac{R^4}{N^2}, \frac{1}{L^3} \frac{R^4}{N^2} + \frac{NR^2}{M}\right). \quad (10.8)$$

First we show that matrices with

$$C \ll \frac{NR^2}{M} \asymp \frac{Q}{P} \quad (10.9)$$

can be treated as Type 1 matrices. By (7.2), A and D are bounded, and (7.3) is

$$\theta A + \varphi D = 1 + \theta\varphi \quad (10.10)$$

with $\theta = q_1/q$, $\varphi = a_1/a$. Since (3.3) holds, θ and φ are close to 1 if B_{20} is large in (10.3). If B_{20} is sufficiently large, then

$$|\theta - 1|, |\varphi - 1| \leq \frac{1}{2} \max(1, |A|, |D|),$$

and (10.10) gives $A + D = 2$. For some integers b, c, d the magic matrix has the form

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 + bcd & -bd^2 \\ bc^2 & 1 - bcd \end{pmatrix}.$$

Triangular matrices have already been treated as Type 2, so b, c and d are bounded non-zero integers, and C and D are also bounded. By (7.2) we have $P \asymp Q$, and then (10.8) implies that $C \ll P/Q$. The constant B_{20} and the constant implied in the definition of Type 1 matrices are at our disposal. Thus we can always arrange that $C \gg NR^2/M$ for Type 3 matrices, and drop the last term in (10.8), recovering (7.12). For each choice of z and z_1 we get the estimate (7.13) as in § 7.

The total number of coincidences between minor arcs with $Q \leq q < 2Q$ has now been estimated as

$$\begin{aligned} &\ll VI \frac{R^4}{Q^4} PQ(1 + \Delta_2 \Delta_4^{2/3} J) + VI^2 \left(\frac{R}{Q}\right)^{2/3} (\Delta_2 PQ)^{1/3} \left(\frac{P}{Q} + \frac{Q}{P}\right) \\ &\quad + VI^2 \frac{R^2}{Q^2} \Delta_2 \Delta_4^{2/3} (P^2 + Q^2) \log N + I^2 \frac{R^2}{Q^2} \Delta_1 \Delta_2 \frac{P^2 Q^2}{V^{2/3}}. \end{aligned}$$

When (10.7) holds, then we can choose V by (7.14) to get the estimate

$$\begin{aligned} &\ll I \frac{R^4}{Q^4} PQ \left(\frac{1}{\Delta_4} + \frac{\Delta_2}{\Delta_4^{1/3}} J\right) + I^2 \frac{R^2}{Q^2} \frac{\Delta_2}{\Delta_4^{1/3}} (P^2 + Q^2) \log N + I^2 \frac{R^2}{Q^2} \Delta_1 \Delta_2 \Delta_4^{2/3} P^2 Q^2 \\ &\ll \left(\frac{Q}{R}\right)^{2/3} \left(\frac{IM}{R} \left(1 + \left(\frac{R}{N}\right)^{8/3} J\right) + \frac{I^2 M^2 \log N}{N^{11/3} R^{1/3}} + \frac{I^2 R^{11/3} \log N}{N^{5/3}} + \frac{I^2 M^2 R^{14/3}}{N^{20/3}}\right), \quad (10.11) \end{aligned}$$

corresponding to (7.15).

As in § 7, for $V > 4\Delta_1 Q^2$ all magic matrices are upper triangular, and for $P \geq Q$ there is a choice of V satisfying (7.16) for which the total number of coincidences between minor arcs with $Q \leq q < 2Q$ is

$$\begin{aligned} &\ll \frac{R^2}{Q^2} \left(I \Delta_1 PQ^3 + I^2 \left(\frac{Q}{R}\right)^{4/3} \Delta_1 \Delta_2^{1/3} (PQ)^{4/3} + IJ \Delta_1 \Delta_2 \Delta_4^{2/3} PQ^3 \right. \\ &\quad \left. + I^2 \Delta_1 \Delta_2 \Delta_4^{2/3} P^2 Q^2 \log N \right) \\ &\ll \left(\frac{Q}{R}\right)^{2/3} \left(\frac{IMR^4}{N^3} + \frac{I^2 (MR^2)^{4/3}}{N^4} + \frac{I^2 M^2 R^{14/3}}{N^{20/3}} \left(\log N + \frac{NR^2 J}{M I} \right) \right). \quad (10.12) \end{aligned}$$

Similarly when $P \leq Q/B_{17}$, then all magic matrices are lower triangular for $V > B_{18}\Delta_1 P^2$, and there is a choice of V satisfying (7.18) for which the total number of coincidences between minor arcs with $Q \leq q < 2Q$ is

$$\begin{aligned} & \ll \frac{R^2}{Q^2} \left(I\Delta_1 P^3 Q + I^2 \left(\frac{Q}{R} \right)^{4/3} \Delta_1 \Delta_2^{1/3} (PQ)^{4/3} + IJ\Delta_1 \Delta_2 \Delta_4^{2/3} P^3 Q \right. \\ & \quad \left. + I^2 \Delta_1 \Delta_2 \Delta_4^{2/3} P^2 Q^2 \log N \right) \\ & \ll \left(\frac{Q}{R} \right)^{2/3} \left(\frac{IM^3}{N^5} + \frac{I^2(MR^2)^{4/3}}{N^4} + \frac{I^2 M^2 R^{14/3}}{N^{20/3}} \left(\log N + \frac{M}{NR^2} \frac{J}{I} \right) \right). \end{aligned} \quad (10.13)$$

11. Exponential sums with a parameter

We can now prove a sharper version of Theorem 2 of [10]. As with Theorem 2 of [8], the conditions of the theorem are those of [10], and only the conclusion is new.

THEOREM 2. *Suppose that, for some $\delta > 0$, for $1 - \delta \leq x \leq 2 + \delta$, $0 \leq y \leq 1$, the function $F(x, y)$ is six times differentiable, and for some constants $C_1 \geq 1$, $C_2 > 0$,*

$$|\partial_1^r \partial_2^s F(x, y)| \leq C_1$$

for $2 \leq r \leq 6$, $0 \leq s \leq 2$, $r + s \leq 6$, and

$$1/C_1 \leq |\partial_1^r F(x, y)| \quad (11.1)$$

for $r = 2, 3$, and

$$\left| \partial_1 \frac{\partial_1^{r-1} \partial_2 F(x, y)}{\partial_1^r F(x, y)} \right| \geq C_2 \quad (11.2)$$

for $r = 3$. Suppose also that either Case 1 or Case 2 holds:

Case 1: $M \ll T^{1/2}$ and (11.1) and (11.2) hold for $r = 4$;

Case 2: $M \gg T^{1/2}$ and

$$|3F_{111}^2 - F_{11}F_{1111}| \geq C_3, \quad |\Delta| \geq C_4,$$

for some positive constants C_3, C_4 , where

$$\Delta = \begin{vmatrix} 3F_{111}^2 + 4F_{11}F_{1111} & 3F_{11}F_{111} & F_{11}^2 \\ F_{11111} & F_{1111} & F_{111} \\ F_{11112} & F_{1112} & F_{112} \end{vmatrix}.$$

Let S_i be the sum defined in (10.1) and let $y_1, \dots, y_l$ be real numbers satisfying (10.2). Let $\varepsilon > 0$ be arbitrary. Then in both cases for

$$\frac{T^{8/57}}{T^{11/19}} + \frac{J}{I} \gg \frac{T}{M^2} + \frac{M^2}{T}, \quad (11.3)$$

we have

$$\sum_{i=1}^l |S_i|^5 \ll I^{16/19} M^{5/2} T^{89/114 + \varepsilon} + I^{8/11} J^{3/11} M^{5/2} T^{49/66 + \varepsilon}. \quad (11.4)$$

Also there is a constant C_5 for which in Case 1, with $M \leq T^{1/2}/C_5$,

$$\sum_{i=1}^I |S_i|^5 \ll IM^{5/2} T^{5/6+\varepsilon} E_1^{3/8} \left(1 + \frac{JM^2}{IT}\right)^{1/2}, \quad (11.5)$$

with

$$E_1 = \frac{1}{T^{8/33}} + \frac{T^{2/3}}{M^2} + \left(J + \frac{IT}{M^2}\right)^{-1},$$

and in Case 2, with $M \geq C_5 T^{1/2}$,

$$\sum_{i=1}^I |S_i|^5 \ll IM^{5/2} T^{5/6+\varepsilon} E_2^{3/8} \left(1 + \frac{JT}{IM^2}\right)^{1/2}, \quad (11.6)$$

with

$$E_2 = \frac{1}{T^{8/33}} + \frac{M^2}{T^{4/3}} + \left(J + \frac{IM^2}{T}\right)^{-1}.$$

The constant C_5 and the implied constants depend on δ , ε , C_1 , C_2 , C_3 and C_4 and on the implied constants in the definitions of Cases 1 and 2.

REMARKS. These bounds are rather elaborate. Treating all arcs as major gives

$$\sum_{i=1}^I |S_i|^5 \ll IM^{5/2} T^{5/6+\varepsilon}, \quad (11.7)$$

corresponding to the 'exponent pair bound $(\frac{1}{6}, \frac{2}{3})$ ', for which see [3]. If I is much smaller than J , then Theorem 1 is better than Theorem 2. Otherwise (11.4) improves (11.7) in the exponents both of T and of I .

The value of the constant C_5 does not matter very much, since (11.4) is the best result for M close to $T^{1/2}$.

Proof. The structure of the proof is the same as that of Theorem 1, but we use the results of § 10 in place of those of § 7. If the term $I^2 M^2 R^{14/3} / N^{20/3}$ dominates in (10.11), (10.12) or in (10.13), then

$$S \ll \frac{IM^5 T^\varepsilon}{N^{11/6} R^{2/3}}. \quad (11.8)$$

There is a new case when the term involving J is the second largest term in (10.11). For $J/I \gg T^{1/3}$, the result follows from (11.7). For

$$\frac{T^{1/3}}{B_{21}} \geq \frac{J}{I} \gg \left(\frac{T^8}{I^{33}}\right)^{1/57} + \frac{T}{M^2} + \frac{M^2}{T}, \quad (11.9)$$

we can choose

$$N \asymp (J/I)^{-2/11} M T^{-3/11}, \quad R \asymp (J/I)^{1/11} M T^{-4/11},$$

so that (8.5), (8.6) and (8.7) hold, and (10.7) holds because (10.2) gives $I \leq J + 1$. This case gives the second term in (11.4).

If J/I is smaller than the lower bound in (11.9), but

$$I^{11/38} T^{49/114} \ll M \ll I^{-11/38} T^{65/114}, \quad (11.10)$$

then we choose

$$N \asymp I^{2/19} M T^{-17/57}, \quad R \asymp I^{-1/19} M T^{-20/57}.$$

Then (8.5), (8.6) and (8.7) hold, whilst (10.7) holds when $I \ll T^{8/33}$, which is also the condition for the range (11.9) to be non-empty. This case gives the first term in (11.4).

For $M \leq T^{1/2}/C_5$ we may choose V by (7.16) and use (10.12) in place of (10.11). If T/M^2 is so large that (11.9) and (11.10) are both false, then using (10.12) leads to a better result. For $J/I \ll T/M^2$ and $M \gg I^{1/4} T^{5/12}$, the term M^2/IT dominates in E_1 , and we choose

$$N \asymp I^{1/4} M^{1/2} T^{-1/12}, \quad R \asymp I^{-1/8} M^{5/4} T^{-11/24}.$$

If J^{-1} is the dominant term in E_1 , then we choose

$$N \asymp J^{1/4} M T^{-1/3}, \quad R \asymp J^{-1/8} M T^{-1/3}. \quad (11.11)$$

If either of the first two terms in E_1 dominates, then the choice of N and R is forced by (10.7) or by the condition $V \geq 1$. We cannot neglect the term in J in (10.12), so we must insert a factor $(1 + JM^2/IT)^{1/2}$ in (11.8). If the first term in E_1 dominates, then (10.7) is critical, and we choose

$$N \asymp M T^{-3/11}, \quad R \asymp M T^{-4/11}. \quad (11.12)$$

If the second term in E_1 dominates, then $V \geq 1$ is the critical condition, and we choose

$$N \asymp R^2 \asymp (M^3/T)^{1/2}.$$

In all these four cases (8.5), (8.6), (8.7) and (10.7) hold when $E_1 \leq 1$. If $E_1 > 1$, then (11.5) is a consequence of the simple bound (11.7).

For $M \geq C_5 T^{1/2}$ we may choose V by (7.18) and use (10.13) in place of (10.11). If M^2/T is so large that (11.9) and (11.10) are both false, then using (10.13) leads to a better result. For $J/I \ll M^2/T$ and $M \gg I^{-1/4} T^{7/12}$, the term T/IM^2 dominates in E_2 , and we choose

$$N \asymp I^{1/4} M^{3/2} T^{-7/12}, \quad R \asymp I^{-1/8} M^{3/4} T^{-5/24}.$$

If J^{-1} is the dominant term in E_2 , then we choose N and R by (11.11) as in Case 1. If either of the first two terms in E_2 dominates, then the choice of N and R is forced by (10.7) or by the condition $V \geq 1$. We cannot neglect the term in J in (10.13), so we must insert a factor $(1 + JT/IM^2)^{1/2}$ in (11.8). If the first term in E_2 dominates, then (10.7) is critical, and we choose N and R by (11.12) as in Case 1. If the second term in E_2 dominates, then $V \geq 1$ is the critical condition, and we choose

$$N \asymp M^{1/2}, \quad R \asymp M^{5/4}/T^{1/2}.$$

In all these four cases (8.5), (8.6), (8.7) and (10.7) hold when $E_2 \leq 1$. If $E_2 > 1$, then (11.6) is a consequence of the simple bound (11.7).

12. Further bounds for exponential sums

We start with a brief account of the situation for general exponential sums. Following Fouvry and Iwaniec [2] we call $F(x)$ a monomial function (of one

variable) if $F'(z) = Cz^{-s}$ for some real s and some non-zero constant C (thus $\log z$ counts as a monomial). We say that $F(z)$ satisfies the virial condition if, for some positive s ,

$$F'(z) = Cz^{-s} + G(z),$$

where $G(z)$ is regular on an open set containing the segment $1 \leq x \leq 2$ of the real axis in the z -plane. Here $G(z)$ involves the parameters M and T of (1.1) and tends uniformly to zero as M and T tend to infinity. The virial condition says that $F(z)$ is approximately a monomial function with s greater than zero.

The van der Corput iteration (see [3] or [13]) consists of Step A (Weyl's differencing lemma) which keeps M the same, decreases T and sends functions satisfying the virial condition for s to functions satisfying the virial condition for $s + 1$, and Step B (Poisson summation) which keeps T the same and replaces M by a number about T/M in size, s by $1/s$. For sums in several variables one would like to be able to use Step A in all variables simultaneously. There is no clean result corresponding to the one-dimensional Step A. Steps A and B are iterated, and the final sum is estimated either trivially, or by the method of Bombieri and Iwaniec used in this paper, or, in the special case of a monomial function of several variables, by the large sieve directly as used by Fouvry and Iwaniec in [2].

Theorems 1 and 2 are invariant under Step B. We give as Theorem 3 the result of applying Step A once, then Theorem 2 with the parameter coming from the differencing variable introduced in Step A (see [10] for details). In an iteration Theorem 2 is used at the end of a sequence of Steps A and B. Each Step A introduces a parameter. The parameters from two or more consecutive Steps A may combined. If, in the application of Theorem 2, any parameter runs through a range of length $\gg T^{8/33}$, then we gain the maximum saving that Theorem 2 allows.

THEOREM 3. *Let $F(x)$ be a real function seven times differentiable for $1 - \delta \leq x \leq 2 + \delta$, with*

$$F^{(r)}(x) \neq 0 \tag{12.1}$$

for $r = 3, 4$,

$$\frac{d^2}{dx^2} \log F^{(r)} \neq 0 \tag{12.2}$$

for $r = 3$,

$$3F^{(4)^2} - F^{(3)}F^{(5)} \neq 0,$$

and

$$\begin{vmatrix} 3F^{(4)^2} + 4F^{(3)}F^{(5)} & 3F^{(3)}F^{(4)} & F^{(3)^2} \\ F^{(5)} & F^{(4)} & F^{(3)} \\ F^{(6)} & F^{(5)} & F^{(4)} \end{vmatrix} \neq 0$$

for $1 - \delta \leq x \leq 2 + \delta$. Suppose in (1.1) that $M = T^\alpha$, and T is sufficiently large in terms of δ . If $\alpha \leq \frac{552}{1549}$ ($= 0.3564\dots$), then we also require (12.1) for $r = 5$ and (12.2) for $r = 4$. Then for any $\varepsilon > 0$,

$$S \ll T^{\beta+\varepsilon} \tag{12.3}$$

with

$$\beta = \begin{cases} \frac{1}{146}(13 + 94\alpha) & \text{for } \alpha \leq \frac{87}{275} (=0.3164...), \\ \frac{1}{244}(11 + 191\alpha) & \text{for } \frac{87}{275} \leq \alpha \leq \frac{423}{1295} (=0.3266...), \\ \frac{1}{1282}(89 + 908\alpha) & \text{for } \frac{423}{1295} \leq \alpha \leq \frac{227}{601} (=0.3777...), \\ \frac{1}{280}(29 + 173\alpha) & \text{for } \frac{227}{601} \leq \alpha \leq \frac{12}{31} (=0.3871...), \\ \frac{1}{128}(4 + 103\alpha) & \text{for } \frac{12}{31} \leq \alpha \leq 1. \end{cases}$$

The implied constant factor in (12.3) depends on ε and on the function F .

The deduction of Theorem 3 from Theorem 2 is the same as in [10]. When F satisfies the virial condition, then the hypotheses of Theorem 3 reduce to the non-vanishing of certain polynomials in s . The real roots of these polynomials are all less than or equal to zero.

We summarise further results assuming that $F(x)$ satisfies the virial condition (for each result in the table, the virial condition could be replaced by a finite set of conditions on the derivatives of F). As in Theorem 3, we write $M = T^\alpha$. The bounds take the form $S \ll T^{\beta+\varepsilon}$ for any $\varepsilon > 0$. The implied constant depends on ε , but not on α . In the first column of the table, θ indicates Theorems 1 or 2, $A\theta$ means applying Step A and then Theorem 2.

Rows of the table bracketed together indicate an upper bound with several terms. The last two columns give the range of α for which each term is of use. There are two types of critical values of α . Within a set of bracketed rows we meet values of α at which one term takes over from another as dominant term. These correspond to re-entrant (concave) vertices of the piecewise linear

Steps	Exponent β	from $\alpha =$	to $\alpha =$
θ	$\begin{cases} \frac{1}{60}(4 + 39\alpha) \\ \frac{1}{120}(29 + 42\alpha) \end{cases}$	$\frac{7}{12}$ $\frac{65}{114}$	$\frac{517}{873} = 0.5922...$ $\frac{7}{12} = 0.5833...$
θ	$\frac{1}{370}(89 + 285\alpha)$	$\frac{49}{114}$	$\frac{65}{114} = 0.5702...$
θ	$\begin{cases} \frac{1}{120}(11 + 78\alpha) \\ \frac{1}{60}(13 + 21\alpha) \end{cases}$	$\frac{5}{12}$ $\frac{356}{873}$	$\frac{49}{114} = 0.4298...$ $\frac{5}{12} = 0.4167...$
$A\theta$	$\begin{cases} \frac{1}{128}(4 + 103\alpha) \\ \frac{1}{280}(29 + 173\alpha) \end{cases}$	$\frac{12}{31}$ $\frac{227}{601}$	$\frac{356}{873} = 0.4078...$ $\frac{12}{31} = 0.3871...$
$A\theta$	$\frac{1}{1282}(89 + 908\alpha)$	$\frac{423}{1295}$	$\frac{227}{601} = 0.3777...$
$A\theta$	$\begin{cases} \frac{1}{244}(11 + 191\alpha) \\ \frac{1}{146}(13 + 94\alpha) \end{cases}$	$\frac{87}{275}$ $\frac{1424}{4747}$	$\frac{423}{1295} = 0.3266...$ $\frac{87}{275} = 0.3164...$
$A^2\theta$	$\begin{cases} \frac{1}{264}(4 + 235\alpha) \\ \frac{1}{1614}(49 + 1351\alpha) \\ \frac{1}{600}(29 + 464\alpha) \end{cases}$	$\frac{120}{419}$ $\frac{967}{3428}$ $\frac{199}{716}$	$\frac{1424}{4747} = 0.3000...$ $\frac{120}{419} = 0.2864...$ $\frac{967}{3428} = 0.2821...$
$A^2\theta$	$\frac{1}{2706}(89 + 2243\alpha)$	$\frac{19}{74}$	$\frac{199}{716} = 0.2779...$
$A^2\theta$	$\begin{cases} \frac{1}{492}(11 + 428\alpha) \\ \frac{1}{318}(13 + 253\alpha) \end{cases}$	$\frac{161}{646}$ $\frac{2848}{12173}$	$\frac{19}{74} = 0.2568...$ $\frac{161}{646} = 0.2492...$

polygonal graph of β against α . Between two unbracketed rows we meet values of α at which the argument giving the best estimate changes. These correspond to convex vertices of the polygonal graph.

For values of $\alpha < \frac{1}{4}$ not in the table we apply Step A repeatedly until α comes into range. The parameters from the A steps may be combined as in Theorem 4 of [10] to get an extra saving on those terms in the upper bound of Theorem 2 for which I occurs to a power less than 1; better than the new exponents after a Step A calculated simply using Weyl's Lemma. For values of $\alpha > \frac{1}{2}$ not in the table we apply Step B first.

The exponent in the Lindelöf problem is at most $\sup(\beta - \frac{1}{2}\alpha)$, which occurs for $\frac{49}{114} \leq \alpha \leq \frac{65}{114}$.

Exponential sum bounds are usually given in exponent pair form. To the relation $\beta = (a + b\alpha)/c$ corresponds the exponent pair $(a/c, (a + b)/c)$, or by some authors $(a/c, (a + b)/c - \frac{1}{2})$. These forms are convenient to use when calculating the van der Corput iteration.

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